

Estimate Active Information Storage for Neural Spike Trains

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This thesis is submitted in partial fulfillment of
the requirements for the degree of
Bachelor of Advanced Computing (Honours)

School of Computer Science
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30 May 2025



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Abstract

Understanding how neural systems store and utilize historical information is fundamental to uncovering mechanisms underlying memory and cognition. Active Information Storage (AIS) quantifies the extent to which past information is retained by a process to predict future states (Lizier et al., 2012). However, traditional AIS analyses typically require artificially segmenting continuous neural activity into discrete time intervals, thereby losing precise spike-timing information and introducing systematic biases.

Building upon recent theoretical advances in continuous-time AIS (Spinney and Lizier, 2018), this study presents the first robust implementation of continuous-time AIS estimation for neural spike trains. Our method operates directly on exact spike timings, thus preserving the natural event-based structure of neural activity. We had conducted extensive validation using diverse synthetic datasets, demonstrating the method’s high accuracy and robustness across various event patterns, strongly supporting the significance of precise spike-timing information in AIS analysis.

After validation on synthetic data, we applied this framework to real-world spike recordings from cultured cortical neural networks. Temporal analysis revealed a systematic increase in information storage capacity as networks matured, with growth patterns varying across cultures during development. Spatial analysis demonstrated that information storage initially emerged within localized regions and subsequently expanded across the network, though with significant heterogeneity among different cultures. We also found that the information storage capacity of the neural networks matured earlier than the information transfer capacity, suggesting a sequential maturation model where internal memory develops before network communication.

This study presents the first practical implementation of continuous-time AIS estimation for neural spike trains, providing a novel analytical tool that preserves precise spike-timing information. Our robust estimator enables deeper insights into neural information dynamics and offers a broadly applicable framework for studying information storage across various complex systems and disciplines.

Acknowledgements

I would like to express my sincere gratitude to Dr. Joseph Lizier for his detailed guidance, instruction, and advice throughout the writing of this thesis. His wonderful ideas and insights have been invaluable to my research and academic development.

I am deeply grateful to my family for their unwavering support throughout my university journey. Their encouragement and belief in me have been a constant source of strength.

I would also like to thank all the teachers I have had during my time at university. Each has contributed to my learning and growth in their own unique way.

Finally, I want to thank my girlfriend Olivia for her continuous support and understanding. Her presence and encouragement have made this journey all the more meaningful.

This work would not have been possible without the support and contributions of all those mentioned above.

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CHAPTER 1

Introduction

Complex dynamical systems—from neural circuits and ecological networks to financial markets and social systems—exhibit intricate behaviors emerging from interactions among their constituent elements over time. A fundamental feature of these systems is their ability to process information through three interconnected dimensions: storing information from past states, transferring information between components, and internally modifying information (Lizier, 2014). Understanding these information dynamics and their role in shaping system behavior remains a central challenge within complex systems science.

Neural networks provide a prime example of complex dynamical systems due to their inherently temporal and dynamic nature. Neural activity underpins essential human functions, from cognition to sensory perception, highlighting the importance of comprehending neural information processing. Within neural networks, information is predominantly encoded through neural spike trains—sequences of discrete action potentials occurring at precise yet irregular intervals. These spike trains serve as the fundamental units of neural computation (Dayan and Abbott, 2001). Their sparse, irregular, and history-dependent characteristics make them particularly suited for studying information storage, as the timing and patterns of previous spikes strongly influence future neural activity.

The ability of neural systems to store and utilize historical information is crucial for maintaining coherent activity patterns and generating appropriate behavioral responses (Gerstner and Kistler, 2002). Quantifying how neurons leverage past information provides critical insights into neural coding principles and the underlying mechanisms of memory formation.

Information theory, pioneered by Claude Shannon (Shannon, 1948), offers a rigorous mathematical framework to quantify such information-processing capabilities. Within this theoretical context, Active Information Storage (AIS) measures the extent to which a process retains and utilizes past information to predict its future states (Lizier et al., 2012). Unlike Transfer Entropy, which quantifies information

flow between system components, AIS uniquely characterizes memory and temporal integration within dynamical systems.

However, traditional discrete-time approaches to estimating AIS encounter significant challenges when applied to spike train data. Conventional methods rely on discretizing continuous-time spike events into bins, inherently sacrificing precise timing information and introducing systematic biases (Spinney and Lizier, 2018). These methodological limitations significantly compromise AIS estimation accuracy and consequently our understanding of neural information processing.

A rigorous solution to these challenges emerged with the development of a continuous-time AIS formulation explicitly tailored for event-based data such as neural spike trains (Spinney and Lizier, 2018). This approach eliminates the need for temporal binning by directly leveraging precise spike timings. The practical benefits of continuous-time frameworks have been demonstrated through successful applications, such as the use of continuous-time Transfer Entropy to analyze neural development data (Shorten et al., 2022). Despite theoretical advancements, practical implementation and validation of continuous-time AIS remain underexplored, especially within the context of developing neural networks.

Recent efforts to implement continuous-time information measures (Mijatovic et al., 2022) have revealed a key theoretical limitation: the formula employed differs from the original continuous-time AIS expression, leaving a conceptual gap in aligning practical implementations with foundational principles. This thesis aims to address these critical methodological gaps by developing and rigorously validating a computationally robust continuous-time AIS estimation framework tailored specifically for neural spike train analysis. Furthermore, this framework is applied empirically to investigate neural information storage dynamics during development, thereby offering novel insights into the temporal emergence of intrinsic information storage within neural circuits.

1.1 Contribution

This thesis makes several significant contributions to the field of neural information dynamics and computational neuroscience. First, we develop a robust computational framework that implements the continuous-time AIS formulation specifically for neural spike train analysis. This implementation represents a significant methodological advancement, as it eliminates the need for temporal binning—a common limitation in existing approaches (Spinney and Lizier, 2018)—and directly operates on precise spike timings. Our framework provides the first comprehensive toolset for continuous-time AIS analysis

of neural data, addressing a critical gap in computational neuroscience and building on recent theoretical advances.

Second, we rigorously validate the continuous-time AIS estimator using synthetic datasets representing both memoryless and memory-rich scenarios. This validation demonstrates the estimator’s accuracy and rapid convergence, confirming its reliability and suitability for analyzing real neural data.

Third, we demonstrate the practical utility of our framework through its application to neural development data, providing new insights into the emergence of information storage capabilities in developing networks. This application extends beyond theoretical validation to address fundamental questions in developmental neuroscience, offering a novel quantitative approach to studying neural network maturation. Our analysis framework reveals previously uncharacterized aspects of memory development in neural networks, contributing to our understanding of neural plasticity and development.

Beyond neuroscience, the estimator’s reliance on timestamped events alone makes it immediately transferable to financial transactions, ecological interactions, and social-media—any domain where past events shape future outcomes on irregular timelines—thus opening a general pathway to quantify, compare, and ultimately influence how complex systems store and use information over time.

1.2 Thesis Outline

Chapter 2 provides a comprehensive literature review covering the theoretical foundations of information theory, the development of Active Information Storage from discrete to continuous-time formulations, and current applications in neuroscience. This review identifies critical methodological gaps that motivate the present work.

Chapter 3 presents the mathematical framework and computational implementation of continuous-time AIS estimation for spike trains. The chapter details the theoretical foundations, entropy estimation methods, and practical implementation of the estimator and significance testing.

Chapter 4 validates the proposed estimator through extensive testing on synthetic datasets with known properties. The validation establishes the reliability and accuracy of our continuous-time AIS estimator across different firing patterns and recording conditions.

Chapter 5 applies the validated framework to real neural culture data, investigating how information storage capabilities develop over time. The analysis reveals systematic patterns in memory development and provides novel insights into neural network maturation.

Chapter 6 concludes by summarizing the key contributions, discussing broader implications for neural information processing, and outlining future research directions.

Literature Review / Related Works

This chapter provides a comprehensive review of the theoretical foundations and empirical applications relevant to continuous-time Active Information Storage (AIS) estimation for neural spike trains. We begin with essential information-theoretic concepts that underpin AIS, then examine the development of AIS theory from discrete to continuous-time formulations. Subsequently, we review current applications of AIS in neuroscience across different data modalities. Finally, we identify key research gaps that motivate the present work.

Understanding information processing in complex dynamical systems requires rigorous mathematical frameworks for quantifying how systems store, transfer, and modify information over time. AIS specifically measures how much information from a system's past states contributes to predicting its future behavior, providing insights into memory mechanisms and temporal integration capabilities.

2.1 Information-Theoretic Foundations

Information theory is built upon the concept of entropy and was first introduced by Claude Shannon in his paper "A Mathematical Theory of Communication" (Shannon, 1948). Information theory provides the mathematical framework for quantifying and analyzing uncertainty, randomness, and the transfer of information within complex dynamical systems (Cover and Thomas, 2006). This section introduces essential concepts underpinning the information-dynamics measures explored in this thesis, particularly Active Information Storage (AIS).

2.1.1 Shannon Information Content

Information content, also known as self information, measures the information of a specific outcome x from a discrete random variable X with probability mass function $p(x)$ (Shannon, 1948; MacKay, 2003). Formally, information content is defined as:

$$I(x_i) = \log_b \left(\frac{1}{p(x_i)} \right) = -\log_b p(x_i) \quad (2.1)$$

The logarithmic base b determines the measurement unit. The most common units are bits ($b = 2$) and nats ($b = e$) (Cover and Thomas, 2006). This applies to all the information theoretic measures that follow. Information content is inversely proportional to the probability of the event. As the probability of a particular outcome decreases, the Shannon information content increases, representing greater surprise. For an event with an outcome that has a probability of 1, the information content is 0, which represents there is no surprise in an event that is certain to occur.

2.1.2 Entropy

Entropy generalizes Information Content (Section 2.1.1) by quantifying the average uncertainty or information content across all outcomes of a random variable (Shannon, 1948; Cover and Thomas, 2006). It is computed as the weighted sum of the information content for each possible outcome x_i , with weights corresponding to their probabilities $p(x_i)$:

$$H(X) = \sum_{x_i \in X} p(x_i) \log_2 \left(\frac{1}{p(x_i)} \right) = E[I(X)] \quad (2.2)$$

Entropy is always non-negative. It is zero when the random variable is deterministic—that is, when there is only one possible outcome with probability 1 and no information is needed to describe it. Conversely, entropy reaches its maximum when the distribution is uniform, meaning all outcomes are equally likely (Cover and Thomas, 2006).

2.1.2.1 Cross Entropy

Cross entropy generalizes the concept of entropy to compare two probability distributions defined over the same set of outcomes (Cover and Thomas, 2006). While entropy quantifies the average uncertainty

inherent in a single probability distribution, cross entropy measures the expected number of bits required to encode samples from a true distribution P using a coding scheme optimized for a different distribution Q .

Given two discrete probability distributions $P = \{p(x_i)\}$ (the true distribution) and $Q = \{q(x_i)\}$ (the estimated or model distribution) over the same random variable X , the cross entropy $H(P, Q)$ is defined as:

$$H(P, Q) = - \sum_{x_i} p(x_i) \log_2 q(x_i) = \mathbb{E}_P[-\log_2 q(X)] \quad (2.3)$$

Intuitively, cross entropy quantifies the inefficiency that arises when the coding or modeling of data is based on Q rather than the true distribution P . The closer Q is to P , the lower the cross entropy. In fact, cross entropy is always greater than or equal to the entropy of P , i.e., $H(P, Q) \geq H(P)$, with equality if and only if $Q = P$. This property underpins its widespread use in machine learning as a loss function for classification tasks, where minimizing cross entropy encourages the model distribution Q to approximate the true distribution P as closely as possible.

2.1.2.2 Joint Entropy

The entropy can be applied to two or more discrete random variables. Joint entropy measures the average uncertainty or information content of a set of discrete random variables together (Cover and Thomas, 2006; MacKay, 2003). Formally, for two discrete random variables X and Y with joint probability $p(x, y)$, the joint entropy $H(X, Y)$ is defined as:

$$H(X, Y) = \sum_{x_i \in X, y_i \in Y} p(x_i, y_i) \log_2 \left(\frac{1}{p(x_i, y_i)} \right) \quad (2.4)$$

Joint entropy can be generalized to an arbitrary number of random variables beyond two variables. Similar to entropy, joint entropy is maximized when all joint outcomes are equally likely which represents the greatest uncertainty.

2.1.2.3 Differential Entropy

Previously we mainly focused on the discrete form of entropy (Section 2.1.2). This concept can be naturally extended to continuous random variables using differential entropy (Shannon, 1948; Cover and Thomas, 2006). Formally, for a continuous random variable X with probability density function $f(x)$, the differential entropy is defined as:

$$h(X) = - \int_{S_x} f(x) \log f(x) dx \quad (2.5)$$

where $S_x = \{x : f(x) > 0\}$ denotes the set of all values x for which the probability density function $f(x)$ is nonzero, i.e., the region where the random variable can take values. Unlike discrete entropy, differential entropy can take negative values because probability density functions are not bounded above by 1, especially when the mass is concentrated in a very small region (Cover and Thomas, 2006). In such cases, $-f(x) \log f(x)$ can be negative. It is important to note that differential entropy is not an absolute measure of uncertainty, and its value depends on the coordinate system and units used. It is best understood in comparative settings such as mutual information (Section 2.1.3) which remain non negative.

2.1.3 Mutual Information

Another fundamental measure in information theory is mutual information, also known as information gain (Shannon, 1948; McGill, 1954). Mutual information is the reduction in uncertainty of one variable when the other variable is known. It can be treated as the shared/common information between two variables. The formal definition of mutual information $I(X, Y)$ is the average reduction in uncertainty about one variable X that results from learning the other variable Y or vice versa.

Formally, the mutual information $I(X; Y)$ is defined as the expected reduction in the entropy of one variable due to knowledge of the other (Cover and Thomas, 2006):

$$I(X, Y) = H(X) + H(Y) - H(X, Y) \quad (2.6)$$

This formulation shows that the shared information is the sum of the individual uncertainties minus their joint uncertainty. A helpful way to visualize this is with a Venn diagram: the overlap between the entropies of X and Y represents the mutual information (MacKay, 2003).

Alternatively, mutual information can be expressed in terms of the joint and marginal probabilities (Cover and Thomas, 2006):

$$I(X, Y) = \sum_{x \in X, y \in Y} p(x, y) \log_2 \frac{p(x, y)}{p(x)p(y)} = \sum_{x \in X, y \in Y} p(x, y) \log_2 \frac{p(x|y)}{p(x)} \quad (2.7)$$

This formulation highlights that mutual information compares the joint distribution $p(x, y)$ with the product of the marginal distributions $p(x)p(y)$, which would hold under independence.

Mutual information applies to both discrete and continuous random variables. For the continuous case, analogous to how differential entropy extends discrete entropy, we define mutual information in terms of probability density functions as follows (Cover and Thomas, 2006; Kraskov et al., 2004):

$$I(X, Y) = \int_{x \in X, y \in Y} p(x, y) \ln \frac{p(x, y)}{p(x)p(y)} = \int_{x \in X, y \in Y} p(x, y) \ln \frac{p(x|y)}{p(x)} \quad (2.8)$$

Note that for continuous variables, the logarithm is typically taken in natural units (nats), and integrals replace the summations. Like its discrete counterpart, mutual information in the continuous setting remains non-negative and captures the statistical dependency between variables (Cover and Thomas, 2006).

2.2 Active Information Storage in Discrete Time

Having established the foundational concepts of information theory (Section 2.1)—we now turn to examine how these principles can be applied to understand information processing in dynamical systems. Building upon these mathematical frameworks, we can now explore how complex systems utilize temporal information.

The core objective of Information Theory is to study how information is processed within systems. Generally speaking, information processing can be decomposed into three fundamental components: information storage, which refers to how systems retain information from their past states to influence future behavior; information transfer, which describes how information flows or propagates between different parts of the system; and information modification, which captures how information within the system is combined, processed, or transformed to create new information (Lizier, 2014).

Among these three components, information storage plays a particularly fundamental role in understanding the temporal dynamics of complex systems. The ability of a system to retain and utilize information from its past is crucial for prediction, adaptation, and memory formation—processes that are central to biological, neural, and artificial systems alike. In neuroscience, for instance, understanding how individual neurons and neural networks store temporal information is essential for deciphering mechanisms of learning, memory consolidation, and predictive coding (Friston, 2010). Moreover, information storage provides insights into the intrinsic computational capabilities of systems, revealing how much of their future behavior is determined by their historical states rather than external influences or random fluctuations. This focus on internal memory mechanisms makes AIS particularly valuable for studying autonomous systems and understanding the emergence of complex behaviors from simpler temporal dependencies.

2.2.1 Formal Definition and Intuition

To understand Active Information Storage (AIS), we first introduce the concept of an embedding length k . Here, k represents how many past time steps we look back upon to help predict the next state. For example, if we set $k = 3$, it means we use the current state and the states from the previous two time steps to predict the next one.

Suppose $X = \{X_n\}$ is a discrete-time stochastic process. Given an embedding length k , we define a past state vector:

$$X_n^{(k)} = \{X_n, X_{n-1}, \dots, X_{n-k+1}\} \quad (2.9)$$

The Active Information Storage $A_X^{(k)}$ is then defined as the mutual information (Section 2.1.3) between this past state vector $X_n^{(k)}$ and the next state X_{n+1} :

$$A_X^{(k)} = I(X_n^{(k)}; X_{n+1}) \quad (2.10)$$

Intuitively, this definition measures how much the uncertainty of the next state is reduced once the past k states are known.

AIS can also be expressed in terms of entropy (Section 2.1.2):

$$A_X^{(k)} = H(X_{n+1}) - H(X_{n+1} | X_n^{(k)}) \quad (2.11)$$

This expression indicates how much our initial uncertainty about X_{n+1} decreases once the past k states are known. A substantial reduction (small conditional entropy $H(X_{n+1} | X_n^{(k)})$) implies that the system has strong predictability, indicating high memory capacity, and thus a large AIS value.

From a probabilistic perspective, mutual information can also be represented as the expected value of the log-likelihood ratio:

$$A_X^{(k)} = \left\langle \log_2 \frac{p(x_{n+1} | x_n^{(k)})}{p(x_{n+1})} \right\rangle \quad (2.12)$$

Here, the angle brackets $\langle \cdot \rangle$ indicate averaging (expectation) over all possible combinations of past states $x_n^{(k)}$ and future states x_{n+1} . This formula essentially compares "the probability of a specific future state given knowledge of past states" with "the probability of that state without such knowledge." If knowing the past significantly alters our prediction of the future, the log ratio becomes large, thus increasing the AIS value.

In addition to this global AIS, we can compute a corresponding local AIS value at each specific time point n :

$$a_X(n, k) = \log_2 \frac{p(x_{n+1} | x_n^{(k)})}{p(x_{n+1})} \quad (2.13)$$

Local AIS captures how much information a specific past realization $x_n^{(k)}$ contributes to predicting a specific future state x_{n+1} . In other words, it quantifies how effectively the system's past predicts its immediate future at a particular moment.

By averaging local AIS across all time points, we recover the global AIS:

$$A_X^{(k)} = \langle a_X(n, k) \rangle \quad (2.14)$$

2.2.2 Relations to similar measures

To clarify what sets AIS apart, it is useful to compare it with other closely related information-theoretic measures: Transfer Entropy (TE) and Predictive Information (also known as Excess Entropy). Although all of these measures are grounded in information theory, each one highlights a different aspect of temporal dependence and information flow in complex systems.

Transfer Entropy (TE): Inter-Process Information Flow. Transfer Entropy (TE) was originally proposed as a measure for quantifying directed information flow between stochastic processes (Schreiber, 2000). Unlike AIS, which focuses on how a system's own past informs its future, TE specifically measures how much information flows from one process to another.

Formally, transfer entropy quantifies how much knowing the past of a source process Y helps reduce uncertainty about the future of a target process X , above and beyond what is already explained by X 's own past. The key point is that TE conditions on the past of X to isolate the unique contribution of Y , ensuring that it measures true information transfer rather than effects due to shared inputs or common history. Mathematically, TE is defined as:

$$T_{Y \rightarrow X}(k, l) = I(X_{n+1}; Y_n^{(l)} \mid X_n^{(k)}) \quad (2.15)$$

where $X_n^{(k)} = \{X_n, X_{n-1}, \dots, X_{n-k+1}\}$ and $Y_n^{(l)} = \{Y_n, Y_{n-1}, \dots, Y_{n-l+1}\}$ denote the recent past states of processes X and Y , respectively, with embedding dimensions k and l .

To illustrate the distinction, consider two neurons A and B. AIS for neuron A quantifies how much neuron A's own spiking history helps predict its next spike. In contrast, TE from B to A measures how much neuron B's spiking history provides additional information about A's next spike, beyond what A's own history already tells us. This makes TE particularly valuable for inferring causal relationships and functional connectivity in neural networks.

Predictive Information (Excess Entropy): Long-term Memory. Predictive Information, also known as Excess Entropy (Crutchfield and Young, 1989; Feldman and Crutchfield, 1998), takes a fundamentally different temporal perspective compared to AIS. While AIS focuses on short-term predictive relationships between recent past states and the immediate next state, Predictive Information captures the total amount of information from the *entire* past of a process that constrains its *entire* future.

Formally, Predictive Information is defined as:

$$E(X) = I(X_{past}; X_{future}) = I(\{X_{-\infty}, \dots, X_n\}; \{X_{n+1}, \dots, X_{\infty}\}) \quad (2.16)$$

This measure provides a global characterization of how the complete historical trajectory constrains the future evolution of a system across all timescales. In practical implementations, the infinite past and future are approximated using sufficiently long finite sequences.

To understand this distinction, consider a neuron with complex firing patterns that include both short-term refractory effects and long-term oscillatory behaviors. AIS would capture how the neuron’s firing in the last few milliseconds (refractory period) predicts whether it will fire in the next millisecond, while Predictive Information would capture how the neuron’s entire firing history—including oscillatory patterns spanning seconds or minutes—constrains its future firing patterns over similar long timescales.

The temporal scope distinction is crucial: AIS provides local, short-term memory utilization measuring how k recent states predict the next state, whereas Predictive Information captures global, long-term predictive dependencies spanning from all past to all future states.

2.3 Active Information Storage in Continuous Time

While the discrete-time AIS framework (Section 2.2) provides a solid foundation for analyzing temporal dependencies in systems with regular sampling intervals, many real-world processes—particularly neural spike trains—evolve continuously in time. As established earlier in this chapter, neural spikes occur at precise moments rather than at fixed time steps, and the timing of these events carries crucial information about neural computation and memory. Moreover, manually discretizing such continuous processes through binning can lead to significant information loss and estimation biases. Therefore, developing a rigorous continuous-time extension of AIS is essential for accurately quantifying information storage in neural systems and other natural phenomena that operate in continuous time.

2.3.1 Neural Spike Trains as Point Processes

Neural communication relies on brief electrical impulses called action potentials or spikes that travel along neurons. When we record from a neuron over time, we observe a sequence of these spike

events—this sequence is called a spike train. Think of it like recording the times when a neuron fires during an experiment.

Unlike continuous signals such as audio recordings or temperature measurements, spike trains are fundamentally different: they consist of discrete, non-overlapping events (individual action potentials) that occur at specific moments in time, with potentially long periods of silence between events. This creates a unique data structure that requires specialized mathematical treatment.

Formally, a spike train is the ordered set of spike times $X = \{x_i\}_{i=1}^{N_X}$ observed in the interval $[0, T]$ (Gerstner and Kistler, 2002; Dayan and Abbott, 2001). In simpler terms, if we record from a neuron for T seconds and observe N_X spikes, then X is just a list of the exact times when each spike occurred, arranged in chronological order.

To analyze these spike trains, we often want to count how many spikes have occurred up to any given time. The associated counting process

$$N_X(u) = \sum_{i=1}^{N_X} \mathbf{1}_{\{x_i \leq u\}}, \quad (2.17)$$

returns the total number of spikes that have occurred up to time u (Daley and Vere-Jones, 2003). The indicator function $\mathbf{1}_{\{x_i \leq u\}}$ equals 1 if spike i occurred at or before time u , and 0 otherwise.

Another important characteristic is the timing between consecutive spikes. Local timing is captured by the inter-spike intervals (ISIs) $h_{x_i} = x_i - x_{i-1}$ for $i \geq 2$. These intervals tell us how long the neuron waited between firing events, which contains crucial information about the neuron's internal state and computational properties.

Spike trains display three key properties that make them particularly suitable for continuous-time analysis, rather than treating them as simple binary sequences. First, spiking is generally sparse and irregular: neurons often stay silent for long stretches punctuated by short bursts of firing. Second, spike generation is strongly history-dependent; refractory effects and synaptic integration mean that the likelihood of a new spike depends on recent spikes (Gerstner and Kistler, 2002). For example, immediately after firing, a neuron enters a refractory period where it cannot fire again, making the timing of past spikes crucial for predicting future ones. Third, firing statistics are frequently non-stationary, adapting with behavioural context, stimulus conditions, or long-term plasticity (Churchland and Shenoy, 2007).

Based on these characteristics, it is necessary to model spike trains as continuous-time point processes. This stochastic framework naturally represents discrete event timings along a continuous timeline, enabling more effective capture of the temporal precision and history-dependent characteristics of spike events (Daley and Vere-Jones, 2003).

2.3.2 Inherent Limitations of Discrete-Time Approaches

The most straightforward approach to applying information-theoretic measures to spike trains involves discretization: dividing time into fixed intervals (bins) and converting spike trains into binary sequences that indicate whether a spike occurred in each bin. While conceptually simple, this approach introduces fundamental problems that compromise the analysis of neural spike trains.

Figure 2.1 intuitively illustrates the differences between continuous and discrete methods. This binning procedure introduces several critical drawbacks that are particularly problematic for point processes like neural spike trains.

Information Loss Through Temporal Quantization. The first major limitation is the loss of precise timing information. Consider a neuron that fires twice within a recording period: once at 10.2 ms and again at 10.8 ms. If we use 1 ms bins, both spikes fall into the same bin (the 10-11 ms interval), making them indistinguishable despite occurring 0.6 ms apart. This temporal precision loss is critical because many neural computations depend on spike timing with millisecond or sub-millisecond accuracy.

More fundamentally, as demonstrated by Shorten et al. (2021), binning-based estimators are mathematically consistent—that is, they are guaranteed to converge to the true value as the amount of data increases—only when the underlying process is truly discrete in time. In contrast, for continuous-time processes such as neural spike trains, any choice of finite bin width inevitably results in the loss of precise timing information. As a result, no matter how much data is collected, the estimator will converge to an incorrect value.

Statistical Robustness vs. Temporal Resolution Trade-off. The second major limitation creates a fundamental dilemma in the estimation process. To mitigate the temporal quantization bias, one might attempt to use smaller bin widths Δt . However, this approach triggers the curse of dimensionality: to cover the same temporal span, the history vectors must include many more bins, causing the joint state space to grow exponentially (Shorten et al., 2021). This exponential growth dramatically increases the

amount of data required to populate all possible past-future configurations, making reliable estimation increasingly difficult.

For example, consider estimating AIS for a neuron firing at 10 Hz (a typical rate). With 1 ms bins, only about 1% of bins contain spikes, leaving 99% empty. This sparsity makes it difficult to reliably estimate probability distributions needed for information measures. To achieve statistical robustness, researchers often increase bin sizes to 10-100 ms, but this aggregates many potentially distinct neural events into single bins, potentially masking important temporal patterns. Therefore, discrete methods cannot effectively capture the dynamic characteristics of neurons, and we need to explore continuous-time AIS methods that preserve the natural temporal structure of spike trains.

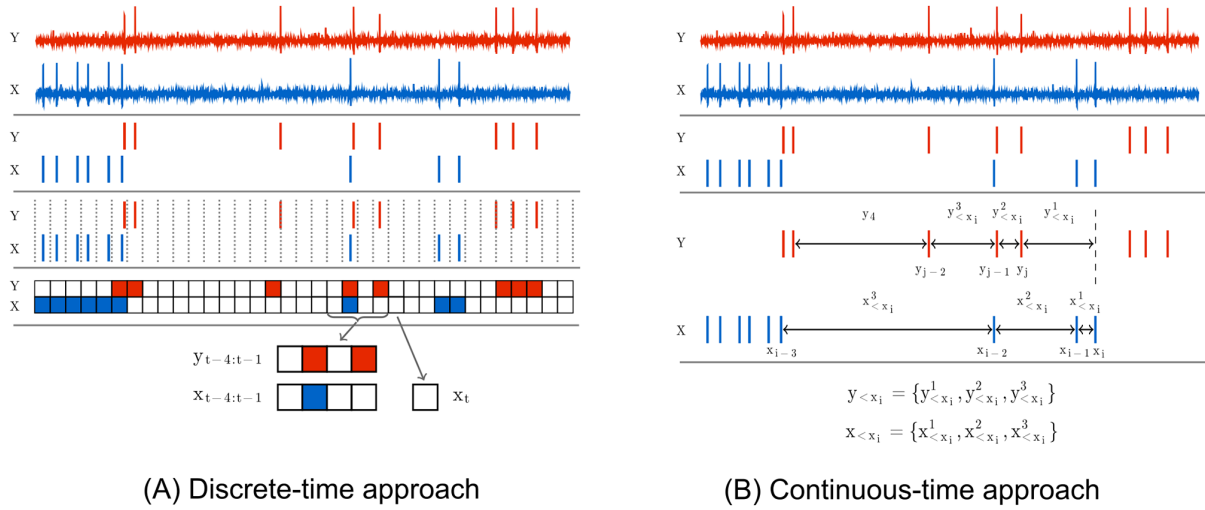


FIGURE 2.1: Diagrams highlighting the differences between discrete and continuous-time estimators. (A) The discrete-time approach divides spike trains into fixed time bins, assigning binary values to indicate spike presence/absence. History embeddings become binary vectors. (B) The continuous-time approach preserves the precise timing of events, constructing history embeddings from raw interspike intervals. Adapted from Shorten et al. (2021).

2.3.3 Continuous-Time AIS Theory

Since the discrete time binning approach (Section 2.3.2) exhibits several intrinsic limitations, it becomes essential to explore methods that naturally treat data as continuous. However, extending the discrete-time Active Information Storage (AIS) formulation to continuous-time processes introduces significant theoretical challenges, as highlighted by Spinney and Lizier (2018). Addressing these challenges requires a careful reconsideration of how AIS fundamentally captures memory and predictability in continuous systems.

2.3.3.1 The Divergence Problem in Continuous Time

A direct extension of the discrete-time AIS formulation to continuous-time processes takes the following form:

$$A_X(\Delta t) = \mathbb{E} \left[\ln \frac{p(x_{t+\Delta t} \mid x_{-\infty:t})}{p(x_{t+\Delta t})} \right] \quad (2.18)$$

where $A_X(\Delta t)$ quantifies the expected information provided by the entire past $x_{-\infty:t}$ about the future state $x_{t+\Delta t}$, for a prediction interval Δt .

However, this direct extension reveals a fundamental mathematical obstacle that must be carefully addressed. Specifically, when the prediction interval becomes arbitrarily small ($\Delta t \rightarrow 0$), the smoothness inherent in continuous trajectories causes fundamental mathematical problems.

To understand why this divergence occurs, consider a simple example: if we know a particle's position at time t , and we want to predict its position at time $t + \Delta t$, where Δt is extremely small. In continuous-time systems, the trajectory is smooth, meaning the position changes continuously without jumps. As Δt approaches zero, the future position $x_{t+\Delta t}$ becomes virtually identical to the current position x_t . This near-perfect predictability from continuity alone causes the conditional probability density $p(x_{t+\Delta t} \mid x_{-\infty:t})$ to approach an infinitely sharp Dirac delta function centered at a value determined by the current state.

Meanwhile, the unconditional probability density $p(x_{t+\Delta t})$, which represents the probability distribution without any historical knowledge, remains finite and smooth across the state space. This creates an extreme imbalance: the numerator becomes infinitely peaked while the denominator stays bounded, leading the AIS expression to diverge to infinity as $\Delta t \rightarrow 0$:

$$A_X(\Delta t) \xrightarrow{\Delta t \rightarrow 0} \infty$$

This divergence occurs because, in the limit of extremely small time intervals, the overwhelming predictive power comes not from genuine historical memory but from the trivial fact that continuous systems cannot change instantaneously. The past value is infinitely close to the future value purely due to trajectory continuity, not due to meaningful information storage.

2.3.3.2 Theoretical Framework and Components

To address this issue, Spinney and Lizier (2018) proposed a rigorous theoretical framework for continuous-time AIS by explicitly decomposing it into two conceptually distinct components:

$$A_X = I_X + \dot{M}_X dt + \mathcal{O}(dt^2) \quad (2.19)$$

The first component, **Instantaneous Predictive Capacity**, I_X , measures the ability to predict future states solely based on the current state. Formally, it is defined as:

$$I_X = \lim_{dt \rightarrow 0} \mathbb{E} \left[\ln \frac{p(x_{t+dt}|x_t)}{p(x_{t+dt})} \right] \quad (2.20)$$

The Instantaneous Predictive Capacity characterizes the instantaneous determinism arising from trajectory smoothness, which inherently causes its divergence in continuous systems. Importantly, this component does not represent genuine system memory but rather reflects the immediate continuity of the state trajectory. Consequently, it is typically disregarded in practical research.

The second component, **Memory Utilisation Rate**, $\dot{M}_X$, captures the incremental predictive information obtained by considering the entire history in addition to the current state. Formally, it is defined as:

$$\dot{M}_X = \lim_{dt \rightarrow 0} \frac{1}{dt} \mathbb{E} \left[\ln \frac{p(x_{t+dt}|x_{-\infty:t})}{p(x_{t+dt}|x_t)} \right] \quad (2.21)$$

The Memory Utilisation Rate genuinely reflects the historical dependence or “memory” of the system. Although it is structurally similar to the Instantaneous Predictive Capacity, this component is constructed as a ratio of two conditional probabilities: one conditioned on the entire past and one conditioned only on the present state. As $dt \rightarrow 0$, both the numerator and denominator in this ratio become increasingly sharply peaked, each approaching a Dirac delta function due to the continuity of the process. If considered separately, these delta-like terms would each diverge, leading to infinite values. However, because both the numerator and denominator diverge in the same way, their ratio effectively “cancels out” the shared divergence—this is the mutual cancellation. As a result, the divergent contributions from trajectory smoothness are removed, and what remains is a finite, well-defined quantity that isolates

the genuine contribution of historical memory to predictability. This mutual cancellation is what ensures that the Memory Utilisation Rate remains finite and stable, even as the prediction interval becomes arbitrarily small.

Intuitively, while the Instantaneous Predictive Capacity quantifies predictability derived solely from the current state (inevitably diverging due to continuous determinism), the Memory Utilisation Rate quantifies the subtle but crucial influence of past information on predicting future states beyond current state knowledge alone. This distinction constitutes one of the foundational insights underpinning continuous-time AIS theory.

2.3.3.3 Memory Utilisation Rate for Neural Spike Trains

Having established the theoretical decomposition of continuous-time AIS into its divergent and finite components, we now turn to Memory Utilisation Rate for neural spike train analysis.

Through detailed mathematical derivations (assuming stationarity and ergodicity, see Spinney and Lizier (2018)), a practical and stable estimation formula for Memory Utilisation Rate applicable to neural spike train data is obtained:

$$\dot{M}_X = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{i=1}^{N(T)} \ln \frac{\lambda(t_i|\text{history})}{\lambda_0} \quad (2.22)$$

Here, t_i represents the occurrence time of the i^{th} spike event for a neuron. The term $\lambda(t_i|\text{history})$ denotes the conditional firing rate, capturing the instantaneous probability of neuron spiking at time t_i , given all preceding spike history. The baseline firing rate λ_0 , on the other hand, characterizes the unconditional average spiking frequency, independent of historical information.

In practical calculations, we calculate this logarithmic ratio at each spike time and take the time average to obtain a stable Memory Utilisation Rate value. However, estimating the history-conditional firing rate involves high-dimensional historical trajectory modeling, and specific methods will be detailed in Chapter 3.

2.3.4 Existing Implementations and Challenges

Recent there have been some implementations of continuous time estimators for information-theoretic measures.

Shorten et al. (2022) successfully implemented continuous-time Transfer Entropy based on Spinney's mathematical framework, employing k -nearest neighbor methods to estimate the underlying probability density functions, together with history embedding techniques. The practical application of this continuous-time Transfer Entropy to neural culture data has yielded remarkable insights into neural network development and organization.

In contrast, Mijatovic et al. (2022) attempted to implement continuous-time AIS but their methodology fundamentally diverged from Spinney and Lizier's theoretical framework. Most critically, their approach abandoned the baseline firing rate comparison that is central to the original formulation (Equation 2.15). Instead of computing the ratio $\frac{\lambda(t_i|\text{history})}{\lambda_0}$ as specified in the theoretical framework, Mijatovic et al. calculated the ratio of conditional firing rates: specifically, the conditional firing rate based on the entire spike history divided by the conditional firing rate based only on the previous single spike.

Currently, there is a lack of robust and theoretically consistent continuous-time AIS implementations. Therefore, subsequent chapters will explore more robust estimation methods to ensure high consistency between theoretical frameworks and practical methods, building upon the successful Transfer Entropy implementations while addressing the problems encountered in AIS estimation.

2.4 Empirical AIS Applications in Neuroscience

Having explored the theoretical foundations of both discrete and continuous-time AIS (Sections 2.2 and 2.3), we now examine how these information-theoretic measures have been applied in practice to understand neural processing across different scales and modalities.

A substantial amount of research has employed information theoretic measures like transfer entropy (TE) and active information storage (AIS) for neural data to understand neural processing. TE is particularly valuable for inferring functional networks and interaction within neural systems. It has been used across different scales of networks from individual neurons to larger scales like brain regions (de la Pava Panche et al., 2019). The directional capability, unlike mutual information (Section 2.1.3) which captures both

directions, allows research to understand the directional dependence of the neurons or brain regions which provides more detailed insight.

Applying AIS to various modalities of neural data offers great potential to complement the insights we can get from TE. AIS with a focus on how much information the past states helps predict the future provides great insights into the memory utilisation within the neurons or groups of neurons. Combined with the application of TE, which quantifies information flow across neurons, we can analyse the neural processing better by considering both internal memory stored within the neurons and also the external information transfer between the neurons. Note that internal memory also incorporates the information transfer through feedback loops which makes AIS more powerful to detect these dependencies (Spinney and Lizier, 2018). There is still a decent amount of research that has successfully applied AIS to different neural data modalities and has gained significant insights into brain functioning and self-predictive dynamics.

Moreover, AIS has been used to analyse the characteristics of different states in functional networks. It is widely believed that the brain works in two states: segregated states, in which brain regions work more independently, and integrated states in which the brain regions have more coordinated communication. In order to understand the behaviour divergence of neurons in these two states, it is essential to understand how they behave in memory utilisation to better gain an understanding on the property of these states.

Segregated states, marked by independent regional processing, exhibit higher AIS due to self-predictive dynamics, while integrated states show lower AIS as information is shared across networks (Li et al., 2019). This aligns with predictive coding frameworks, where segregation prioritizes internal priors, where memory of the past is largely used, while integration emphasizes sensory input therefore utilising less past memory (Friston, 2010).

Active Information Storage (AIS) has proven to be a particularly sensitive tool for detecting disease-related loss of "neural memory." When AIS is computed from resting-state BOLD time-series in drug-naïve Parkinson's disease (PD) patients, it is found to be significantly reduced in the primary motor cortex. This reduction indicates that the cortico-basal-ganglia loop can no longer maintain the stable, self-generated dynamics characteristic of healthy brains. In contrast, traditional rs-fMRI metrics—such as the fractional Amplitude of Low-Frequency Fluctuations (fALFF), which only measure slow-wave power—often reveal only subtle or inconsistent group differences. AIS, by quantifying how well each

voxel's current activity can be predicted from its own recent past, provides a much clearer distinction. In PD, this predictability is markedly diminished, exposing a loss of intrinsic information storage that conventional connectivity or power-based metrics frequently miss (Zou et al., 2008; Puche Sarmiento et al., 2020).

Beyond diagnostic insights, In face or house recognition tasks, researchers using magnetoencephalography (MEG) recordings observed increased Active Information Storage (AIS) in brain regions specifically associated with processing the presented category of visual stimuli. For example, during face recognition, the fusiform face area (FFA), a region strongly linked to face processing, exhibited significantly elevated AIS, reflecting the activation and sustained maintenance of prior knowledge about faces. Further research demonstrated that AIS values from these category-specific brain areas could accurately decode the stimulus category (faces or houses) at the single-trial level, highlighting AIS's capacity to reflect and predict specific perceptual content within neural memory patterns (Brodski-Guerniero et al., 2017). These findings align well with predictive coding theories, which propose that the brain maintains prior knowledge to generate internal predictions about upcoming sensory information (Friston, 2010).

Finally, in studies using electroencephalography (EEG) Active Information Storage (AIS) has been utilized to analyze dynamic variations in brain activity over time. Researchers examining fluctuations of AIS have observed that the brain's capacity for information storage closely tracks the cardiac cycle (Fouragnan et al., 2022; Valenza et al., 2016). Specifically, the AIS level in the brain dynamically adjusts with different phases of the heartbeat, indicating that neural information processing and storage differ according to cardiac rhythms. These dynamic changes uncovered by AIS highlight a nuanced interplay between brain and heart activity, making AIS a powerful measure for capturing and quantifying the time-varying dynamics of neural information processing.

While these macro-scale applications demonstrate AIS's utility for understanding neural information processing at the systems level, a critical gap remains in applying continuous-time AIS to the fundamental computational units of neural networks: individual spike trains. The discrete-time approaches used in most existing work rely on temporal binning, which, as established in Section 2.3.2, introduces systematic biases and loses the precise timing information that is crucial for understanding neural computation. Despite the theoretical advances in continuous-time AIS formulation, practical implementations for spike train analysis remain limited, creating a significant disconnect between the available theoretical framework and its empirical application to the very data type for which it was originally developed.

2.5 Research opportunity / Problem statement

The comprehensive review presented in the preceding sections reveals both significant theoretical advances and critical gaps in the current state of continuous-time information-theoretic analysis of neural systems.

The previous sections have highlighted critical gaps in the application of information-theoretic measures to continuous-time neural data, particularly spike trains (Section 2.3.1). While the theoretical framework for continuous-time AIS exists through the formulation of Spinney and Lizier (2018), there remains a significant disconnect between theory and application, especially for point processes like neural spike trains.

By addressing these gaps, this research will advance both the theoretical understanding and practical analysis of how complex systems encode and utilize temporal information, particularly in continuous-time processes where traditional discrete analyses may be insufficient or misleading.

The primary objective of this thesis is threefold. First, we aim to implement a computational framework for continuous-time AIS specifically designed for neural spike train analysis. This implementation will bridge the gap between the theoretical formalism and practical neuroinformatics applications by creating robust algorithms that operate directly on the temporal structure of spike trains without resorting to artificial binning procedures.

Second, we will undertake a comprehensive validation of the continuous-time AIS estimator using synthetic data with known ground-truth information storage properties. This validation phase is crucial as it will establish the statistical robustness of the implementation across varying firing patterns and recording conditions, thereby ensuring the reliability and accuracy of our methodological framework.

Finally, we will investigate information storage dynamics in neural cultures using continuous-time AIS analysis. This application will enable us to analyze how information storage properties evolve during neuronal growth and network formation, providing valuable insights into how neurons progressively develop their capacity to utilize temporal information as they mature and form connections. Through this systematic approach, our research will advance both the theoretical understanding and practical analysis of how complex systems encode and utilize temporal information, particularly in continuous-time processes where traditional discrete analyses may be insufficient or misleading.

Continuous AIS Estimation

In this chapter, we present our computational framework for continuous-time Active Information Storage (AIS) estimation in neural spike trains. Building upon the theoretical foundation established in Chapter 2, we develop practical methods for implementing the continuous-time AIS formulation introduced by Spinney and Lizier (2018). Our derivation and estimator construction closely follow the approach of Shorten et al. (2021), who provided a practical entropy-based reformulation for continuous-time information-theoretic measures in neural data. We begin by reformulating the continuous-time AIS definition (Eq. 3.1) into an entropy estimation problem suitable for empirical computation. We then present the construction of history embeddings, the k-Nearest Neighbor (kNN) differential-entropy estimation method, and significance assessment through surrogate generation.

3.1 Reformulating Continuous-Time AIS as an Entropy Estimation Problem

We begin by recalling the definition of continuous-time Active Information Storage (AIS) for spike trains, as introduced in Chapter 2:

$$\dot{A}_X = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{i=1}^{N_X} \ln \frac{\lambda_X(x_i | X_{<x_i})}{\bar{\lambda}_X}, \quad \text{with } \bar{\lambda}_X = \frac{N_X}{T}, \quad (3.1)$$

where $\lambda_X(x_i | X_{<x_i})$ is the conditional intensity (i.e., the instantaneous firing rate) of the process X at time x_i given its past $X_{<x_i}$, and $\bar{\lambda}_X$ is the average firing rate.

However, directly estimating the conditional intensity function $\lambda_X(x_i | X_{<x_i})$ is difficult in practice, mainly because the spike history can be high-dimensional and available data is limited. To make AIS estimation tractable, we seek an equivalent formulation that is more amenable to empirical computation.

3.1.1 From Conditional Intensities to Probability Ratios

The key insight for reformulation is to express the conditional intensity in terms of probabilities over infinitesimal time intervals, which will allow us to leverage well-established entropy estimation techniques. This approach transforms the problem from estimating continuous-valued intensity functions to estimating probability density ratios.

Specifically, the conditional intensity at time t can be written as:

$$\lambda_{X,t} = \lim_{\Delta t \rightarrow 0} \frac{p_t(\Delta N_X(t) = 1)}{\Delta t}, \quad (3.2)$$

where $\Delta N_X(t) = N_X(t + \Delta t) - N_X(t) \in \{0, 1\}$ indicates whether a spike occurs in the infinitesimal interval $[t, t + \Delta t)$, and p_t is the probability conditioned on the history up to time t .

This relationship allows us to rewrite the original AIS definition in Eq. (3.1) in terms of probabilities. Substituting the intensity expression and taking the limit as $\Delta t \rightarrow 0$, we obtain an ensemble expectation over spike events:

$$\dot{A}_X = \bar{\lambda}_X \lim_{\Delta t \rightarrow 0} \mathbb{E}_{p_X} \left[\ln \frac{p_U(\Delta N_X(t) = 1 \mid X_{<x_i})}{p_U(\Delta N_X(t) = 1)} \right] \quad (3.3)$$

Here, p_U represents probabilities under a uniform reference measure over all times, independent of spike occurrence.

3.1.2 Applying Bayes' Theorem to Eliminate Intensity Dependencies

The next critical step involves manipulating the conditional probability in the numerator of Eq. (3.3) to eliminate explicit dependence on spike occurrence probabilities. This transformation is essential because it will reveal the underlying structure as a ratio of history densities.

By Bayes' theorem, we can decompose the conditional probability as:

$$p_U(\Delta N_X(t) = 1 \mid X_{<x_i}) = \frac{p_U(X_{<x_i} \mid \Delta N_X(t) = 1) \cdot p_U(\Delta N_X(t) = 1)}{p_U(X_{<x_i})}$$

Substituting this decomposition into Eq. (3.3), we observe that the terms involving $p_U(\Delta N_X(t) = 1)$ in both numerator and denominator cancel exactly:

$$\dot{A}_X = \bar{\lambda}_X \mathbb{E}_{p_X} \left[\ln \frac{p_U(X_{<x_i} \mid \Delta N_X(t) = 1)}{p_U(X_{<x_i})} \right] \quad (3.4)$$

Now, recognizing that $p_U(X_{<x_i} \mid \Delta N_X(t) = 1)$ is precisely the density of histories at spike times (which we denote as $p_X(X_{<x_i})$), we arrive at:

$$\dot{A}_X = \bar{\lambda}_X \mathbb{E}_{p_X} \left[\ln \frac{p_X(X_{<x_i})}{p_U(X_{<x_i})} \right] \quad (3.5)$$

Here, p_X is the distribution of histories at spike times, and p_U is the distribution of histories at arbitrary (reference) times, independent of spikes.

3.1.3 Final Reformulation as Entropy Estimation

The final expression in Eq. (3.5) shows that continuous-time AIS can be computed as the difference between two quantities: the cross-entropy between the spike-time and reference-time history distributions, and the differential entropy of the spike-time history distribution, both scaled by the mean firing rate. Explicitly,

$$\dot{A}_X = \bar{\lambda}_X [-H(p_X) + H_{p_U}(p_X)] \quad (3.6)$$

where $H(p_X)$ is the differential entropy of the spike-time history distribution, and $H_{p_U}(p_X)$ is the cross-entropy between the spike-time and reference-time history distributions.

This reformulation is central to our computational approach: it reduces the challenge of estimating conditional intensity functions to the more tractable problem of estimating differential entropy and cross-entropy of spike history distributions. Our goal now is to accurately estimate these entropy and cross-entropy terms.

3.2 History Embedding Construction

The entropy terms $H(p_X)$ and $H_{p_U}(p_X)$ in Eq. (3.6) involve probability densities over spike histories, which are inherently infinite-dimensional objects. The complete history $X_{<x_i}$ of a spike train up to time x_i encompasses all previous spike times, creating a complex, variable-length representation that cannot be directly used with standard entropy estimation techniques. To make the problem computationally tractable, we must construct finite-dimensional representations of these histories that capture the essential temporal dependencies while remaining amenable to empirical density estimation.

To estimate continuous-time AIS in practice, we need to evaluate the differential entropy and cross-entropy terms in Eq. (3.6). This requires empirical approximations of two probability densities: the

density of process history at target events (spikes) and the density of process history at random times. These densities exist in a potentially high-dimensional space defined by the past activity of the process.

Before entropy estimation, we construct fixed-length embeddings of spike history to represent the influence of the past on future spiking. These embeddings serve as data points sampled from the respective probability densities p_X and p_U that appear in our AIS reformulation.

The first type, **event-time embeddings** ($\mathcal{X}_E$), are constructed from spike times as:

$$X_{<x_i} = [h_{x_i}, h_{x_{i-1}}, \dots, h_{x_{i-d+1}}], \quad (3.7)$$

capturing the most recent d inter-spike intervals prior to a spike. These embeddings sample from p_X in the entropy formulation and represent the distribution of spike history patterns that precede spikes.

The second type, **non-event embeddings** ($\mathcal{X}_U$), are constructed at random, non-spike times u_j as:

$$X_{<u_j} = [h_{u_j}, h_{x_i}, \dots, h_{x_{i-d+2}}], \quad h_{u_j} = u_j - x_i, \quad (3.8)$$

representing past spike intervals at non-spike moments. These embeddings sample from p_U in the entropy formulation and characterize the background distribution of history patterns regardless of future spiking.

Once constructed, the embeddings form matrices $\mathcal{X}_E \in \mathbb{R}^{N_E \times d}$ and $\mathcal{X}_U \in \mathbb{R}^{N_U \times d}$ for spike and non-spike data, respectively.

3.3 k-Nearest Neighbor (kNN) Differential-Entropy Estimation

Why we use a kNN entropy estimator. Following the continuous-time transfer-entropy study of Shorten et al. (2021), we use the Kozachenko–Leonenko k -nearest-neighbour (kNN) estimator for differential entropy. We keep this choice for three practical reasons. *First*, the estimator is non-parametric: under mild assumptions it is essentially unbiased and consistent, and it does not require us to guess the shape of the underlying density (Kozachenko and Leonenko, 1987). *Second*, because it looks only at distances between data points, its bias and variance grow more slowly with dimension than kernel or plug-in methods, which is handy for the moderate embedding sizes we work with ($d \leq 10$) (Kraskov

et al., 2004). *Third*, using the same estimator as Shorten *et al.* keeps our AIS pipeline directly comparable with the existing continuous-time TE literature, letting us study information storage and transfer within one consistent, non-parametric framework.

Constructing kNN estimators for differential entropies. Following the derivation in Shorten et al. (2021), consider an unknown probability distribution $\mu(x)$ for $x \in \mathbb{R}^d$. Given a set $\mathcal{X}$ of N_X points drawn from μ , we need to estimate the differential entropy. After applying the constant-density assumption within small neighborhoods and appropriate bias correction as detailed in Shorten et al. (2021), we arrive at the Kozachenko-Leonenko estimator:

$$\hat{H}_{\text{KL}}(\mathcal{X}) = \ln(N_X - 1) - \psi(k) + \ln c_d + \frac{d}{N_X} \sum_{i=1}^{N_X} \ln \varepsilon(k, x_i, \mathcal{X}), \quad (3.9)$$

This estimator has been proven to be consistent under appropriate conditions (Kozachenko and Leonenko, 1987), where d is the dimensionality of the embedding space (i.e., the number of inter-spike intervals in our history embeddings), ψ is the digamma function, c_d is the volume of the d -dimensional unit ball under the chosen norm, and $\varepsilon(k, x_i, \mathcal{X})$ is the distance to the k -th nearest neighbor of x_i in the set $\mathcal{X}$.

Cross-entropy estimation. To estimate cross-entropy between two distributions, we follow the approach in Shorten et al. (2021). Given two sets of samples $\mathcal{X}$ and $\mathcal{Y}$ drawn from distributions μ and β respectively, the cross-entropy can be estimated using a similar k-nearest neighbor approach, but with an important difference: distances are measured from points in one set to their nearest neighbors in the other set. The final cross-entropy estimator is:

$$\hat{H}_{\beta, \text{KL}}(\mathcal{X}) = -\psi(k) + \ln N_Y + \ln c_d + \frac{d}{N_X} \sum_{i=1}^{N_X} \ln \varepsilon(k, x_i, \mathcal{Y}) \quad (3.10)$$

where $\varepsilon(k, x_i, \mathcal{Y})$ is the distance from point x_i to its k -th nearest neighbor in set $\mathcal{Y}$.

kNN Estimator with same k values. With both the entropy and cross-entropy estimators in hand, we can now construct our estimator for active information storage. Applying these estimators to our AIS entropy formulation in Eq. (3.6), we use $\hat{H}_{\text{KL}}(\mathcal{X}_E)$ to estimate $H(p_X)$ and $\hat{H}_{p_U, \text{KL}}(\mathcal{X}_E)$ to estimate $H_{p_U}(p_X)$. Combining these terms and simplifying yields the final AIS estimator:

$$\hat{A}_X = \bar{\lambda}_X \left[\ln \frac{N_U}{N_E - 1} + \frac{d}{N_E} \sum_{i=1}^{N_E} \ln \frac{\varepsilon(k, x_i, \mathcal{X}_U)}{\varepsilon(k, x_i, \mathcal{X}_E)} \right], \quad (3.11)$$

Generalized kNN Estimator with different k values. Previous work, including the continuous-time transfer entropy estimator of Shorten et al. (2021), has found that sharing the same or similar radii between entropy terms in kNN-based estimators can reduce overall bias, as components of the individual biases may cancel each other. However, when the entropy and cross-entropy terms involve datasets of different sizes (N_E vs N_U) or different local density characteristics, using the same k value for both terms may result in substantially different neighborhood radii, potentially limiting the bias reduction benefits.

To ensure similar radii between the entropy and cross-entropy estimations while maintaining optimal neighborhood sizes for each dataset, we need to use a generalized kNN estimator that allows different k values (k_E and k_U) for the two terms. By carefully selecting these different k values, we can achieve comparable neighborhood radii despite differences in dataset characteristics, thereby maximizing the bias cancellation effects observed in previous work. This motivates the development of our generalized AIS estimator that can accommodate different neighbor counts for the entropy and cross-entropy components.

The generalized form of our AIS estimator is:

$$\hat{A}_X = \bar{\lambda}_X \left[\underbrace{\psi(k_U) - \psi(k_E)}_{\text{digamma bias correction}} + \underbrace{\ln \frac{N_U}{N_E - 1}}_{\text{sample-size correction}} + \underbrace{\frac{d}{N_E} \sum_{i=1}^{N_E} \ln \frac{\varepsilon(k_U, x_i, \mathcal{X}_U)}{\varepsilon(k_E, x_i, \mathcal{X}_E)}}_{\text{ratio of } k\text{-NN radii (local density terms)}} \right]. \quad (3.12)$$

Here: $\psi(k_U) - \psi(k_E)$ restores the individual digamma corrections when $k_E \neq k_U$. The $\ln(N_U/(N_E - 1))$ term is unchanged from the equal- k case. The final term sums the log-ratio of the two sets of k -NN radii, reflecting the difference between the event-time and background local densities. Equation (3.12) reduces to Eq. (3.11) when $k_E = k_U$.

This generalized form provides additional flexibility when estimating AIS in cases where different neighborhood sizes might better capture the local structure of the event-time and background distributions.

3.4 Implementation

3.4.1 Algorithmic Implementation

Algorithm 1 presents the practical implementation of the generalized continuous-time AIS estimator, directly adapting the adaptive neighborhood approach developed by Shorten et al. (2021) for continuous-time transfer entropy estimation. Following their methodology, we apply their proven algorithmic framework to our AIS entropy formulation established in Eq. (3.12).

The algorithm employs the adaptive neighborhood selection mechanism introduced by Shorten et al. (2021) (lines 3-7), which addresses the challenge of ensuring comparable neighborhood radii between entropy and cross-entropy terms when dealing with datasets of different sizes and local density characteristics. Rather than using fixed k values that may yield substantially different radii, the approach first computes the radius to the k_{glob} -th neighbor for each event-time embedding point in both the event set ($\mathcal{X}_E$) and background set ($\mathcal{X}_U$). The maximum of these two radii is selected as a common neighborhood size, ensuring consistent spatial scales for both entropy components. The effective neighbor counts k_E^* and k_U^* are then determined by counting all neighbors within this unified radius in each respective set.

This established adaptive approach maintains the mathematical structure of Eq. (3.12) while preserving the bias cancellation properties demonstrated by Shorten et al. (2021) for kNN-based information-theoretic estimators. The final computation (line 12) directly implements our generalized AIS formula.

3.4.2 Software Framework

We implemented our continuous-time AIS estimator within the Java Information Dynamics Toolkit (JIDT) framework (version 1.6.1) (Lizier, 2014), following the architectural patterns established by Shorten et al. (2021) for continuous-time Transfer Entropy estimation. This ensures compatibility with existing information-theoretic analysis tools and facilitates integrated analysis of both AIS and Transfer Entropy within a unified computational framework.

The implementation utilizes JIDT's k-d tree data structures for efficient nearest-neighbor searches, providing logarithmic-time complexity for k-NN entropy estimation queries. The central component is a continuous-time AIS calculator class that handles history embedding construction for both spike-triggered and random-time patterns, along with surrogate generation and statistical testing capabilities for significance assessment.

Algorithm 1 Continuous-time AIS estimator (with global neighbour count k_{glob})

Require: $\mathcal{X}_E = \{\mathbf{x}_{<x_i}\}_{i=1}^{N_E}$ {event-time embeddings}
Require: $\mathcal{X}_U = \{\mathbf{x}_{<u_j}\}_{j=1}^{N_U}$ {background embeddings}
Require: $\bar{\lambda}_X$ {mean firing rate}
Require: d {embedding dimension}
Require: k_{glob} {global # of neighbours}
Ensure: $\hat{A}_X$
1: $\text{sum} \leftarrow 0$
2: **for** $i = 1, \dots, N_E$ **do**
3: $r_E \leftarrow \text{FINDKTHNN}(k_{\text{glob}}, \mathbf{x}_{<x_i}, \mathcal{X}_E)$
4: $r_U \leftarrow \text{FINDKTHNN}(k_{\text{glob}}, \mathbf{x}_{<x_i}, \mathcal{X}_U)$
5: $r_{\text{max}} \leftarrow \max(r_E, r_U)$
6: $k_E^* \leftarrow \text{COUNTNNS}(\mathbf{x}_{<x_i}, r_{\text{max}}, \mathcal{X}_E)$
7: $k_U^* \leftarrow \text{COUNTNNS}(\mathbf{x}_{<x_i}, r_{\text{max}}, \mathcal{X}_U)$
8: $\varepsilon_E \leftarrow 2 \text{FINDKTHNN}(k_E^*, \mathbf{x}_{<x_i}, \mathcal{X}_E)$
9: $\varepsilon_U \leftarrow 2 \text{FINDKTHNN}(k_U^*, \mathbf{x}_{<x_i}, \mathcal{X}_U)$
10: $\text{sum} \leftarrow \text{sum} + \ln(\varepsilon_U) - \ln(\varepsilon_E) + \psi(k_U^*) - \psi(k_E^*)$
11: **end for**
12: $\hat{A}_X \leftarrow \bar{\lambda}_X \left[\ln \frac{N_U}{N_E - 1} + \frac{d}{N_E} \text{sum} \right]$
13: **return** $\hat{A}_X$

The calculator integrates seamlessly with JIDT’s existing information-theoretic estimators and supports parameter configuration for embedding dimensions and k-NN searches. The source code and documentation are available as part of the JIDT package, ensuring long-term maintenance and integration with the broader information-theoretic analysis ecosystem.

3.5 Significance Assessment and Surrogate Generation for AIS

Determining whether observed Active Information Storage (AIS) genuinely represents temporal memory rather than chance-level fluctuations necessitates rigorous significance testing. Our working hypothesis states that neural spike trains exhibit structured temporal dependencies, which should manifest as AIS values significantly greater than those derived from appropriate null models. As analytic expressions for the sampling distribution of AIS using k-nearest-neighbor estimators are unavailable, we employed surrogate methods to empirically establish a suitable null distribution.

To test the hypothesis of genuine temporal memory, the null hypothesis we seek to reject asserts that the current spike timing is statistically independent of its past spiking history. In other words, under the null hypothesis, the joint probability distribution factorizes such that future spike occurrence times

are independent of previous spike events. Proper surrogate generation methods must preserve key statistical characteristics—particularly overall spike counts, firing rates, and recording durations—while eliminating all temporal dependencies and causal or history-dependent structures within the spike train.

We constructed surrogate spike trains by randomly redistributing the original spike times throughout the recording period. This preserves the total number of spikes and recording duration while completely eliminating any temporal patterns or sequential dependencies in the original data.

To assess statistical significance, we employed a surrogate-based Monte-Carlo approach. For each empirical spike train, we generated multiple surrogate datasets (e.g., $M = 100$ surrogates per original spike train in the experiments presented later). AIS was recalculated for each surrogate dataset, strictly maintaining identical embedding parameters and k-nearest neighbor settings used in the empirical AIS estimation. The AIS values computed from the surrogate spike trains constitute an empirical null distribution.

The observed AIS value A_{obs} from the original spike train was then compared directly against this surrogate distribution. A p-value was calculated according to:

$$p = \frac{1 + \#\{A_{\text{surr}} \geq A_{\text{obs}}\}}{1 + M}, \quad (3.13)$$

where A_{surr} are surrogate-derived AIS values. Significance was declared if the observed AIS significantly exceeded the surrogate baseline ($p < 0.05$). This approach provides a robust, non-parametric statistical test for genuine temporal memory within spike trains.

In summary, our surrogate-based significance testing approach systematically preserves essential spike-train statistical characteristics while rigorously dismantling temporal structure, thereby providing a robust empirical benchmark against which to validate the observed Active Information Storage in neural spike trains.

3.6 Conclusion

In this chapter, we have presented a comprehensive framework for continuous-time Active Information Storage (AIS) estimation in neural spike trains. We reformulated the AIS definition as an entropy estimation problem suitable for empirical computation, and developed practical methods for implementing the

continuous-time AIS formulation. Additionally, we introduced rigorous significance testing procedures and surrogate generation techniques to assess whether observed AIS values reflect genuine temporal memory rather than chance-level fluctuations.

In the next chapter, we will validate the performance of the proposed estimator on synthetic data which contain memorless and memory rich neural data.

Synthetic Data Analysis

The following chapter presents a series of validation experiments using synthetic spike train data. This approach allows us to establish expected values and assess the goodness of our Active Information Storage (AIS) estimation technique by comparing empirical results against known theoretical values. Through systematic analysis of memory-free, refractory, and bursting processes, we demonstrate the accuracy and convergence properties of our continuous-time AIS estimator.

4.1 Poisson Spike Trains

The most critical validation step involves examining a Poisson spiking process. In a Poisson process, each spike occurs independently of all previous spikes, with a constant probability per unit time. This means that the probability of a spike occurring in any interval depends only on the length of the interval, not on the timing or number of previous spikes—there is no memory of the past. As a result, the theoretical ground truth of the Active Information Storage (AIS) is exactly zero, reflecting the complete absence of historical dependence in the process.

4.1.1 Experiments

To probe finite-sample behaviour we simulated spike trains of four different lengths— 10^2 , 10^3 , 10^4 , and 10^5 events. Alongside each spike train, we generated a corresponding set of uniformly distributed "background" time-points, varying the ratio of background points to spikes (N_U/N_X) as detailed in our methodology (Chapter 3). These null samples provide the reference distribution against which spike-time embeddings are compared. Local entropies were estimated with a k-nearest-neighbour approach that uses the Chebyshev (maximum-norm) distance; we repeated the analysis for two neighbourhood sizes, $k = 1$ and $k = 5$, in order to expose the usual bias–variance trade-off. Each parameter combination was replicated 200 times, giving bootstrap confidence envelopes for every learning curve.

4.1.2 Results

The convergence curves in Fig. 4.1 demonstrates that our estimator reliably collapses onto the theoretical AIS of 0, providing a clear numerical confirmation of both consistency and vanishing bias for a memory-free Poisson process. As the number of observed spikes grows, the standard deviation bands tightens and the mean estimate hugs the zero line, showing that additional data only sharpens—not shifts—the result. A closer look at the side-by-side panels reveals the impact of the neighbourhood size in the k-NN step: with $k = 5$ the trajectory reaches the plateau an order of magnitude sooner than with $k = 1$. This improved performance with $k = 5$ stems from a substantial reduction in variance, as averaging over five neighbours stabilises local density estimates, while the associated increase in bias is negligible at this scale.

When we increase the embedding dimension—effectively considering a longer history by including additional past intervals—the estimator shows a slight negative bias, resulting in a mild underestimation of the AIS. However, this bias and its associated variance steadily diminish as the spike count grows, becoming negligible once the dataset surpasses approximately 10^3 spikes. This behavior aligns with expectations for high-dimensional k-nearest-neighbor estimators, confirming that sufficient data substantially mitigates both bias and variance.

4.2 Refractory-Process Data

After establishing that our estimator yields correct results for memory-free (Poisson) spike trains, we next turn to a process that introduces a simple, biologically relevant form of memory: the Poisson process with an absolute refractory period. In this model, after each spike, there is a fixed time window called the refractory period—during which no new spikes can occur. This mimics the physiological property of real neurons, which are unable to fire again immediately after a spike.

Spinney and Lizier (2018) derived the analytic solution for the Active Information Storage (AIS) rate in this process, providing an exact formula for a Poisson process with absolute refractoriness. The process is parameterized by the baseline firing rate μ (in spikes per second), which determines how frequently spikes would occur in the absence of refractoriness, and the refractory period Δ_x (in seconds), which sets the minimum interval between consecutive spikes.

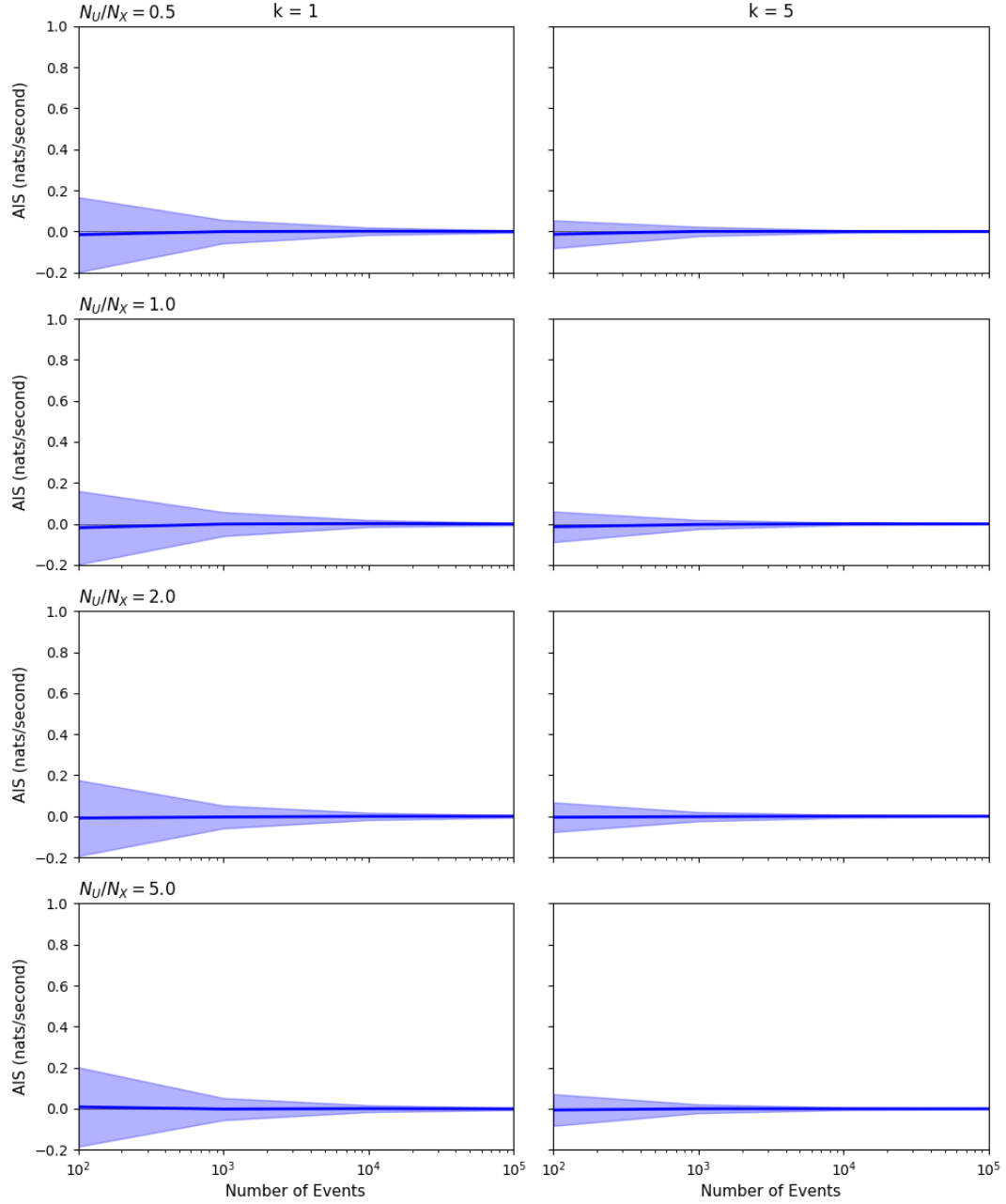


FIGURE 4.1: Convergence of AIS estimates for Poisson spike trains. Left panel shows results for $k = 1$ and right panel for $k = 5$. The dashed line represents the theoretical value of zero. Shaded regions indicate standard deviation across 200 trials.

The theoretical AIS rate for this process is given by:

$$\dot{M}_X = \frac{\mu}{1 + \mu\Delta_x} \ln(1 + \mu\Delta_x) \quad (4.1)$$

This formula quantifies how the introduction of a refractory window increases the memory in the spike train, as longer refractory periods make the timing of future spikes more predictable given the recent past.

To validate our estimator, we systematically varied the refractory period Δ_x across a range of values typical for biological neurons, and compared our empirical AIS estimates to the analytic solution above. This allowed us to directly assess how well our estimator tracks the true AIS as the process transitions from memoryless (Poisson) to increasingly memory-rich dynamics.

4.2.1 Varying Refractory Period

In our first experiment on refractory-period processes, we fixed the spike rate at 100 Hz and varied the refractory period systematically from 0 ms to 20 ms. Each AIS estimate was computed over a fixed recording duration of 10 s, averaging results across 100 independent trials to ensure statistical robustness. As depicted in Figure 4.2, our AIS estimator shows very close agreement with the theoretical solution. Although there is a minor but consistent underestimation of the theoretical AIS at larger refractory intervals (particularly beyond 10 ms), the discrepancy remains small, with percentage errors around only 2%. The AIS values increase monotonically with longer refractory periods, plateauing at approximately 10 ms. This behavior aligns closely with theoretical expectations and confirms that our estimator accurately captures active memory induced by refractory dynamics with minimal bias, thus validating its effectiveness for analyzing realistic neuronal spike processes.

4.2.2 Convergence Analysis

In our second experiment, we assessed the convergence speed of our AIS estimator by varying the number of spikes while keeping the refractory period fixed at 11 ms and the spike rate at 100 Hz. Figure 4.3 demonstrates clearly that our estimator rapidly converges to the theoretical AIS value, achieving stable estimates after as few as 200 spikes. Even with a minimal sample size of around 50 spikes, the estimator provides a reasonable approximation, although it initially exhibits a higher percentage error ($\sim 10\%$). As the spike count increases beyond 200, the percentage error quickly reduces, eventually stabilizing around just 1%. This indicates not only the estimator's accuracy but also its efficiency in reliably capturing active memory, even under relatively sparse sampling conditions. These results confirm that our method provides robust AIS estimates, quickly converging to theoretical expectations with minimal data.

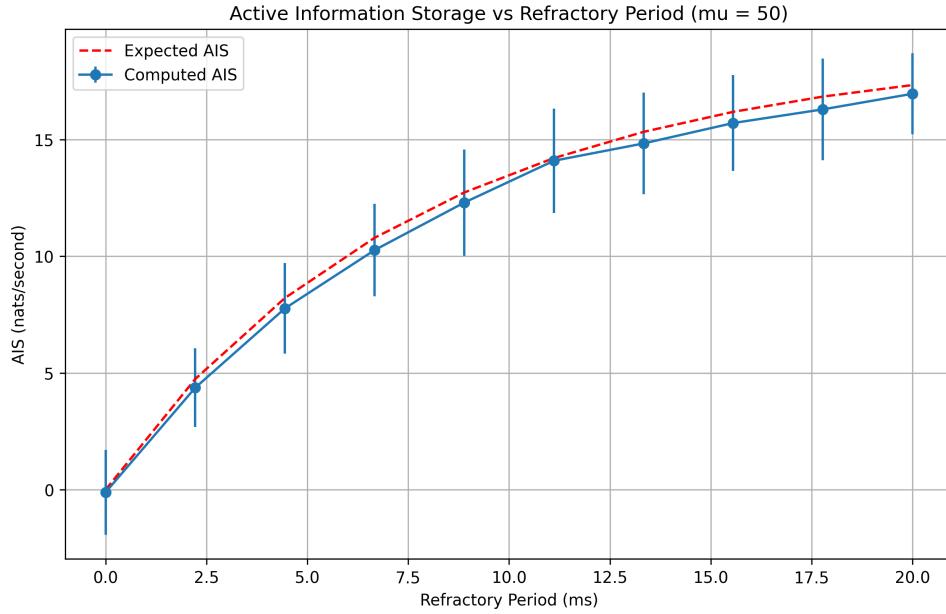


FIGURE 4.2: AIS estimates for refractory-period processes as a function of refractory duration. Blue points represent empirical estimates, while the red curve shows the theoretical value.

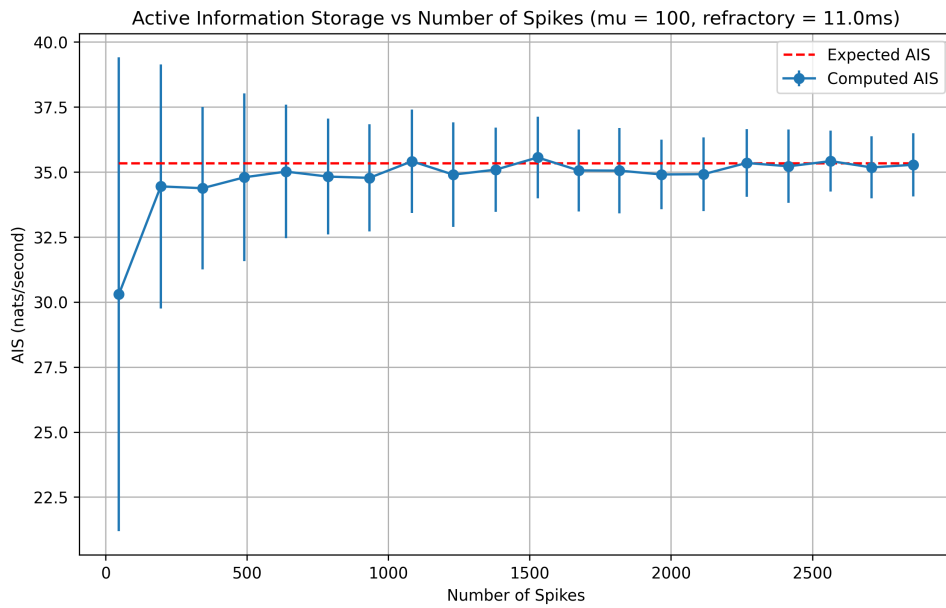


FIGURE 4.3: Convergence of AIS estimates for refractory-period process as a function of spike count. The dashed horizontal line indicates the theoretical value. Error bars represent the standard deviation across trials.

4.3 Bursting Data

Having validated our continuous-time AIS estimator against refractory processes with known theoretical values, we now turn to a more complex but biologically prevalent firing pattern: neuronal bursting. Bursting behavior—where neurons fire in brief, high-frequency episodes separated by quiescent periods—is commonly observed across many neural systems, from cortical pyramidal cells to thalamic relay neurons. Unfortunately, unlike the refractory process case, no analytic solution exists for the AIS of bursting spike trains, making this a crucial test of our estimator’s ability to handle realistic neural dynamics where ground truth is unavailable.

To address this challenge, we first introduce a simplified burst-silent model that captures the essential temporal structure of bursting neurons. We then derive a theoretical approximation for the expected AIS rate based on the statistical properties of this model. Finally, we apply our continuous-time estimator to synthetic bursting data to assess how closely our empirical estimates align with the theoretical approximation, thereby evaluating the estimator’s performance on complex, memory-rich spike patterns typical of real neural activity.

Neuronal firing often comes in bursts—brief packets of several spikes separated by quiescent stretches. To mimic this motif, we construct a simple burst–silent model where each cycle consists of:

- a burst window of length a , during which spikes arrive as a homogeneous Poisson process with rate λ (i.e., the spiking rate is λ only within the bursting window)
- a silent window, whose duration is drawn from an exponential distribution with mean b (i.e., the silent period is itself a Poisson process with expected value b), during which no spikes can occur

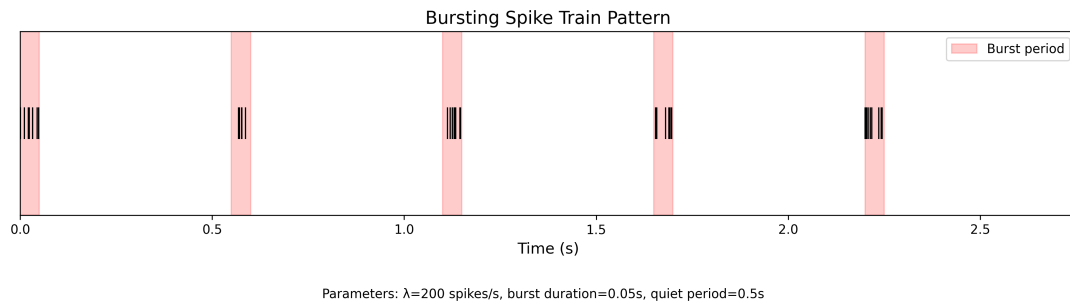


FIGURE 4.4: Illustrative raster plot of a bursting spike train. Here, the silent window is set to fixed intervals for visualization purposes: bursts of duration 0.05 s are separated by fixed silent periods of 0.5 s.

4.3.1 Theoretical Approximation of AIS Rate

As shown in Chapter 3, Eq. (3.1), the Active Information Storage (AIS) rate for a stationary point process is:

$$\dot{A}_X = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{i=1}^{N_X} \ln \frac{\lambda_X(x_i | X_{<x_i})}{\bar{\lambda}_X},$$

where $\bar{\lambda}_X = N_X/T$ is the mean firing rate.

Consider a simplified neuronal spike-generation model characterized by alternating cycles of bursting and silence. Each cycle consists of two distinct intervals: a bursting period of duration a , during which spikes occur following a homogeneous Poisson process at a constant firing rate λ , and a silent period of duration b , during which no spikes occur. These burst-and-silence cycles repeat indefinitely, forming a predictable, rhythmic neuronal activity pattern.

4.3.1.1 Step 1: Unconditional Firing Rate

To calculate the unconditional (long-run) firing rate, we denote the mean firing rate as $\bar{\lambda}_X$. Since spikes only occur during the burst period, the expected number of spikes per cycle is $a\lambda$, and the total cycle duration is $a + b$. Thus,

$$\bar{\lambda}_X = \frac{a\lambda}{a + b}.$$

4.3.1.2 Step 2: Conditional Intensity Approximation (First-order History)

To approximate the conditional intensity $\lambda_X(x_i | X_{<x_i})$ for each spike, we use a first-order history (i.e., only the previous inter-spike interval). There are two cases:

- **First spike in each burst:** The previous interval is the silent period of duration b , so the conditional intensity for this spike is approximately

$$\lambda_X^{(1)} \approx \frac{1}{b}.$$

- **Subsequent spikes within the same burst:** These spikes are separated by intervals of approximately $1/\lambda$, so the conditional intensity is simply

$$\lambda_X^{(\text{burst})} = \lambda.$$

4.3.1.3 Step 3: Calculating Log-Ratio Contributions

Given these two distinct conditional intensities, we compute the associated log-ratio terms, which appear within the AIS estimation formula. For clarity, we define:

- **Log-ratio for the first spike after silence:**

$$\ell_1 = \ln \left(\frac{\lambda_{\text{cond}}^{(1)}}{\lambda_0} \right) = \ln \left(\frac{\frac{1}{b}}{\frac{a\lambda}{a+b}} \right) = \ln \left(\frac{a+b}{a\lambda b} \right).$$

- **Log-ratio for subsequent in-burst spikes:**

$$\ell_b = \ln \left(\frac{\lambda_{\text{cond}}^{(\text{burst})}}{\lambda_0} \right) = \ln \left(\frac{\lambda}{\frac{a\lambda}{a+b}} \right) = \ln \left(\frac{a+b}{a} \right).$$

4.3.1.4 Step 4: Accounting for Number of Spikes and Deriving the AIS Formula

We now explicitly account for how many times each of these log-ratio terms occurs within a typical cycle. Because spikes occur as a Poisson process at rate λ during a burst of length a , the expected number of spikes per burst is $a\lambda$. Of these spikes, exactly one is the first spike that ends the silent period, leaving an expected number of subsequent spikes equal to $a\lambda - 1$.

Summing the contributions of all spikes in one representative cycle, we have:

- **First spike:** contributes ℓ_1 once.
- **Remaining spikes:** each contributes ℓ_b , summing to $(a\lambda - 1)\ell_b$.

Thus, the AIS (Memory Utilization Rate, $\dot{M}_X$) averaged per spike is obtained by dividing the sum by the total number of spikes per cycle, $a\lambda$:

$$\dot{M}_X = \frac{\ell_1 + (a\lambda - 1)\ell_b}{a\lambda}.$$

Substituting the derived expressions for ℓ_1 and ℓ_b , we explicitly obtain the approximate AIS rate:

$$\boxed{\dot{M}_X = \frac{\ln \left(\frac{a+b}{a\lambda b} \right) + (a\lambda - 1) \ln \left(\frac{a+b}{a} \right)}{a\lambda}} \quad (4.2)$$

Having derived an analytic approximation for the AIS rate in the bursting process, we now turn to evaluating how well our estimator recovers this theoretical value in practice. In the following, we compare the analytic prediction to empirical estimates across a range of firing rates and sample sizes.

4.3.2 Firing Rate Variation

In the first bursting scenario (burst duration = 0.05 s, quiet interval = 0.5 s), our continuous-time AIS estimator tracks the theoretical curve very closely across the full range of spike rates, as shown in Figure 4.5. The relative error falls steadily as the within-burst firing rate λ increases, dropping from 13.8% at $\lambda = 50 \text{ spikes s}^{-1}$ to about 3% at $\lambda = 1000 \text{ spikes s}^{-1}$. This monotonic decrease confirms that the estimator becomes more accurate at higher firing rates while consistently remaining within a few percent of the expected value.

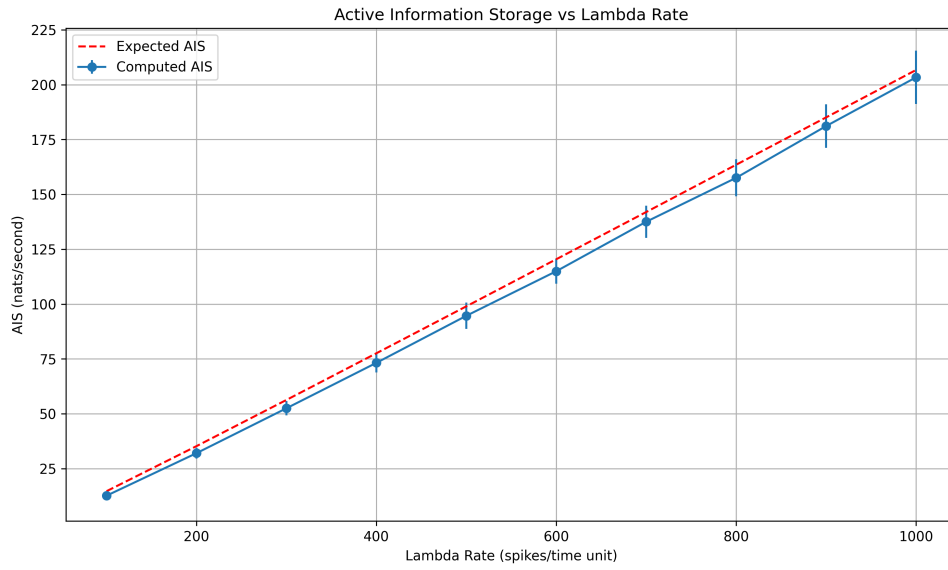


FIGURE 4.5: AIS estimates for bursting data as a function of within-burst firing rate. The theoretical prediction (red curve) closely matches our empirical estimates (blue points). Error bars show standard deviation across trials.

4.3.3 Sample Size Effects

In the second experiment, the within-burst firing rate was held fixed at $\lambda = 200 \text{ spikes s}^{-1}$, while the burst ($a = 0.05 \text{ s}$) and silent ($b = 0.5 \text{ s}$) intervals remained unchanged; only the number of burst–silent cycles—and therefore the total recording time—was increased from 10 to 1000. As Figure 4.6 shows, the sampling variance of the estimator collapses rapidly as more data is accumulated, indicating that

the estimator becomes increasingly precise with larger sample sizes. However, it is notable that the mean estimate drifts downward and eventually stabilises around 31–32 nats s⁻¹, which is approximately 8–10% below the analytic benchmark of 32.5 nats s⁻¹.

Although there is a 10% error observed here, this can be largely attributed to the limited number of spikes within each burst—since with 200 cycles and a burst duration of 0.05 s, there are only about 10 spikes per burst on average. Increasing the within-burst spike count (by raising the firing rate or extending the burst duration) reduces this error, as confirmed by our earlier results. We believe that the remaining discrepancy is primarily due to the limitations of our analytic approximation for the AIS in the bursting process, rather than a deficiency of the estimator itself. The approximation does not fully capture the true memory structure when the number of spikes per burst is small. Thus, the observed bias is best understood as a consequence of the analytic formula’s assumptions, not a fundamental problem with the estimator.

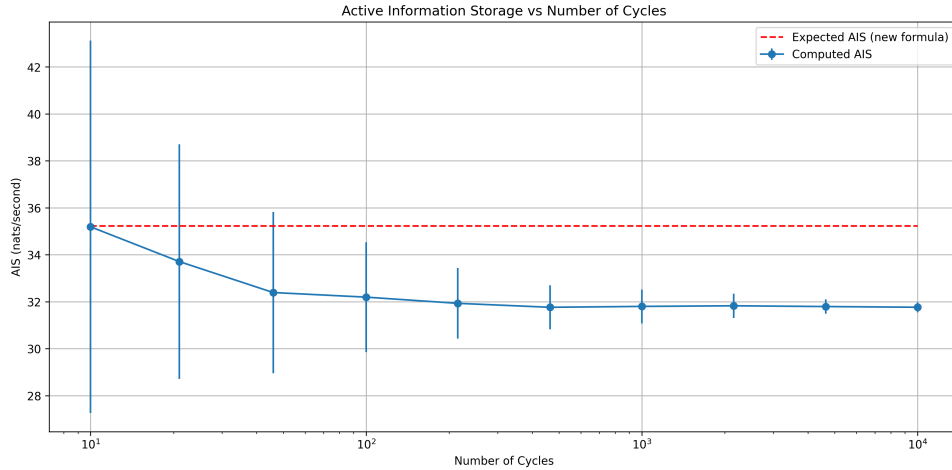


FIGURE 4.6: Convergence of AIS estimates for bursting data as a function of the number of burst-silent cycles. The dashed horizontal line indicates the theoretical value of 32.5 nats s⁻¹. Error bars show standard deviation across trials.

4.4 Conclusion

In this chapter, we presented a series of validation experiments using synthetic spike train data. Our results demonstrate that the estimator achieves high accuracy and converges rapidly to theoretical values across a range of biologically relevant scenarios, including memoryless, refractory, and bursting processes. These findings confirm both the consistency and efficiency of our approach, providing strong evidence that the estimator is robust and reliable for practical applications.

Real-world Data: Neural Culturing Data

Having validated our continuous-time Active Information Storage (AIS) estimator on synthetic data, this study now applies the estimator to real-world neural data. Using AIS, we addressed a fundamental question in neuroscience: How do developing neural networks acquire and utilize memory? Understanding memory utilization throughout neural development is essential as it reveals the maturation trajectory of neurons, allowing the identification of critical transitional phases during neural network formation.

Neural cultures grown on micro-electrode arrays (MEAs) serve as an ideal experimental platform to explore this question. Such cultures enable the study of network maturation under controlled conditions, minimizing external interference and enhancing reproducibility, while simultaneously preserving the intricate dynamic characteristics of living neural networks.

Previous studies employing Transfer Entropy (TE) have identified intriguing patterns of information flow in these cultures, particularly highlighting the early stabilization of network structure, localized communication, and stage-dependent functional integration (Shorten et al., 2022). These prior findings have established essential benchmarks for neuronal communication patterns, but the temporal evolution of internal memory—how individual neurons develop the ability to leverage their past states—remains largely unexplored. We hypothesize that internal memory utilization, quantified by AIS values, progressively increases throughout neural network development, likely exhibiting critical transition points corresponding to key developmental stages.

By systematically analyzing AIS across multiple neural cultures at successive developmental stages, days In Vitro 4-33(DIV 4–33), this study set out to explore three central aspects of information storage: (1) the overall increase of AIS as an indicator of global maturation; (2) the spatial emergence and distribution of AIS hotspots; and (3) the temporal stabilization of AIS patterns.

5.1 Overview of Neural Cultures

Dissociated embryonic rat cortical cells were cultured at high density (2,000–2,500 cells/ μL) on 8×8 micro-electrode arrays (MEAs) featuring 59 active electrodes spaced 200 μm apart, and maintained for several weeks in vitro (DIV 4–33) (Wagenaar et al., 2006; Chiappalone et al., 2006). Rat cortical cells are widely used due to their physiological similarity to human neurons and their relatively rapid maturation in vitro (about 3–4 weeks), making them an established model for systematic studies of neural network development.

Each 59-electrode array recorded extracellular action potentials at 25 kHz, capturing spontaneous bursting activities. Most recordings consisted of randomly sampled half-hour-long recordings, supplemented by continuous overnight sessions. This combined approach provided sufficient data density for accurate information-theoretic estimation and ensured comprehensive temporal coverage of developmental trajectories.

For this study, only the overnight sessions were examined because they offered enough data to estimate AIS. Recordings were analyzed from four neural cultures (1-1, 1-3, 2-2, and 2-5), each covering at least three longitudinal sessions representing early (DIV 4–9), intermediate (DIV 14–22), and late developmental stages (DIV 24–33). Most cultures were recorded on three separate nights, while three (1-3, 2-2, 2-5) have four sessions and one (1-1) has only three. This one(1-1) was selected from those with three sessions because its final recording occurred less than a week after the previous one and was the latest among that group. These recordings come from a publicly available dataset (Wagenaar et al., 2006b; Network, 2021). Comparing AIS values across these days gives snapshots of how each culture’s computational capacity develops.

The cultures analyzed in this study are the same as those used in the Shorten et al. (2022) paper, and, consistent with their findings, these cultures exhibited stable firing and bursting patterns throughout development without signs of decline, meeting the criteria for robust information-theoretic analysis.

5.2 Methods

Active Information Storage (AIS) was estimated using the continuous-time framework developed and validated on synthetic data (Chapter 3). AIS values were computed for each electrode and recording day by embedding inter-spike intervals (ISI) and employing a k-nearest-neighbor (k-NN) entropy estimator.

The estimator operated directly on spike timing data without temporal binning, preserving the original temporal resolution of spike recordings.

Key parameters for AIS estimation were selected based on the comprehensive validation with synthetic data (Chapter 4), where $k=5$ demonstrated optimal bias-variance trade-off compared to $k=1$:

Parameter	Value
History embedding length (ℓ)	1
Nearest neighbor parameter (k)	5
Background sampling ratio (N_U/N_E)	2
Distance metric	Max norm

TABLE 5.1: Parameters used for continuous-time AIS estimation.

The statistical significance of AIS values was assessed using a surrogate data method described in Chapter 3 (Section 3.7). For each electrode on each recording day, 100 surrogate spike trains were generated by randomly redistributing spike times while preserving the original spike count per electrode. This destroyed temporal dependencies, forming a null distribution. AIS values exceeding the 95th percentile of this surrogate distribution were considered statistically significant ($p < 0.05$).

5.3 Results

Our analysis of memory development across multiple developmental stages addresses four fundamental questions: (1) How does memory capacity change during development? (2) What are the detailed evolutionary characteristics of AIS distribution? (3) How is memory distributed spatially across the network and what heterogeneity exists? (4) Does AIS exhibit an stabilization of memory pattern in early periods?

5.3.1 Overall Trend of AIS During Development

Overall, AIS exhibited a clear monotonic increasing trend throughout development: during the earliest stage, the median AIS of all cultures was close to zero; it then gradually rose to approximately $0.11\text{--}3.08$ nats s^{-1} in the intermediate stage, and finally reached $5\text{--}22$ nats s^{-1} in the late stage. For more detailed numerical values, please refer to Table 5.2, which summarizes the median AIS values for each culture and developmental stage. To quantify this trend, we conducted a Wilcoxon signed-rank test (with Bonferroni correction). Among the total of 21 paired-stage comparisons (detailed in Appendix A.1.1),

18 reached statistical significance ($p < 0.00238$); only three intermediate-stage transitions—DIV 9–15 and DIV 15–21 in culture 2-2, and DIV 10–22 in culture 2-5—did not reach significance. These non-significant exceptions are visually marked without asterisks on the paired days in the box plots shown in Figure 5.1.

Culture 1-1	Day 4	Day 14	Day 20	
	0.00	1.86	5.20	
Culture 1-3	Day 5	Day 10	Day 16	Day 24
	0.00	0.11	1.37	5.66
Culture 2-2	Day 9	Day 15	Day 21	Day 33
	3.06	1.14	6.63	22.92
Culture 2-5	Day 4	Day 10	Day 22	Day 28
	0.00	3.08	2.11	14.62

TABLE 5.2: Median AIS values (nats s^{-1}) across developmental stages for each culture. Values demonstrate consistent increases in memory capacity during development.

5.3.2 Detailed Evolution of AIS Distribution Across Developmental Stages

Although the median AIS increased significantly during development overall, the growth rate appeared relatively slow in the early stages. To further analyze this phenomenon, we examined detailed histograms of AIS distributions (see Figure 5.2; each histogram displays significant AIS distributions from 59 electrodes across different cultures and recording days). The histograms also showed the number, mean, and standard deviation of electrodes with significant AIS. From these histograms, we observed a consistently right-skewed distribution of AIS values across cultures at the early stage (DIV 4–5). Most electrodes exhibited very low AIS values, with each culture containing one "pioneer" electrode showing an exceptionally high AIS value. Additionally, the proportion of activated electrodes during this stage was relatively low, only around 15% of the total.

During the intermediate stage, the number of electrodes with significant AIS rapidly increased to about 80%, indicating that more electrodes began to exhibit noticeable intrinsic memory. However, it's important to note that despite the large overall increase in electrode numbers, there were considerable differences among cultures regarding AIS growth trends. The AIS distribution became broader, with more electrodes starting to appear in intermediate-value ranges.

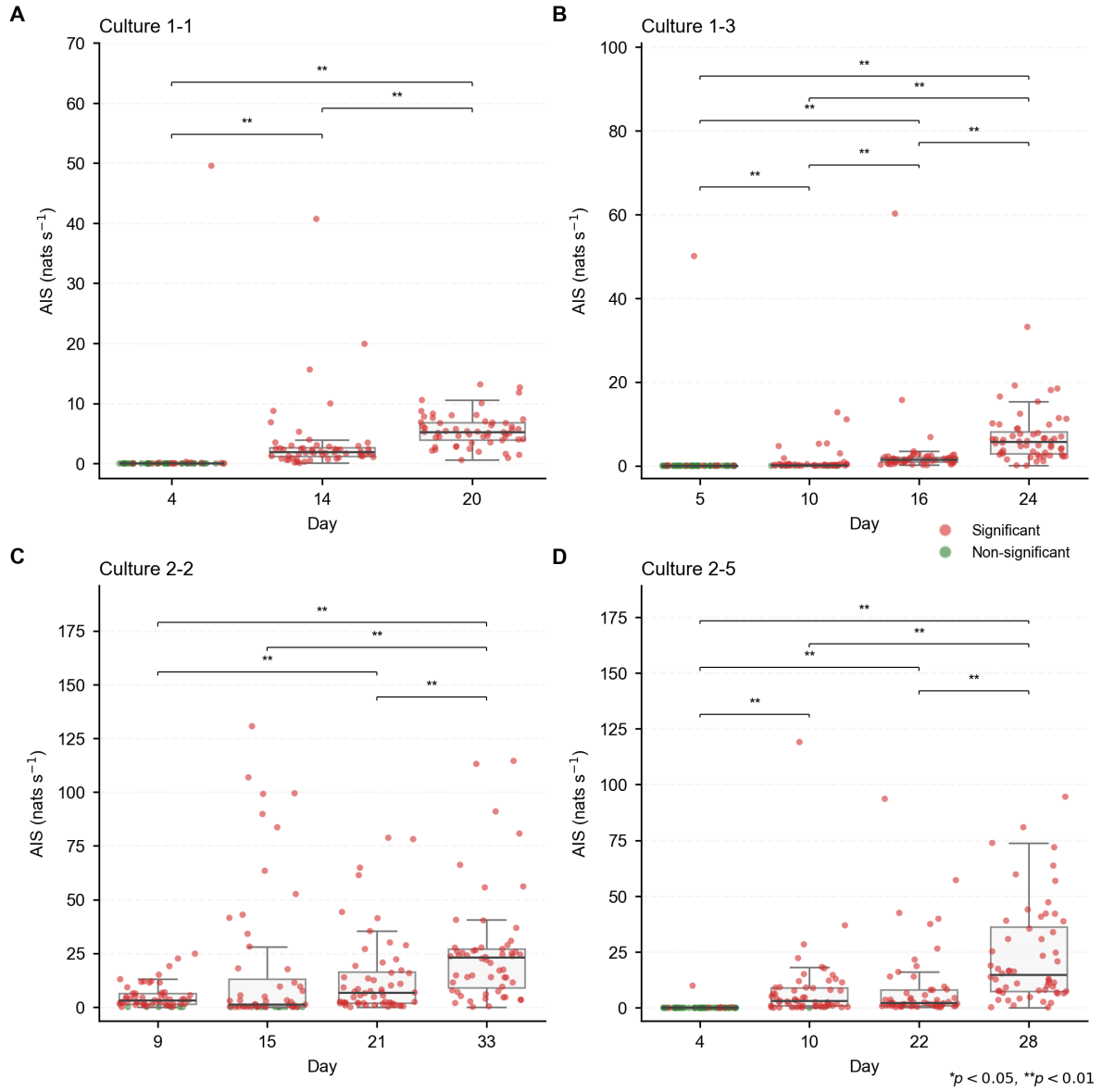


FIGURE 5.1: Developmental progression of AIS distributions across cultures. Red markers denote statistically significant AIS values; connecting lines indicate significant pairwise comparisons.

In the late stage (DIV 16 and later), all cultures reached a state where 56 electrodes (100%) exhibited significant AIS. It is worth mentioning that we excluded three electrodes that consistently failed to exhibit significant AIS throughout all recorded time points; these electrodes were not considered in subsequent analyses. At this stage, the AIS distribution shifted further toward higher values, notably increasing the number of electrodes within the medium-value range, reflecting a robust maturation of intrinsic network memory across the entire network.

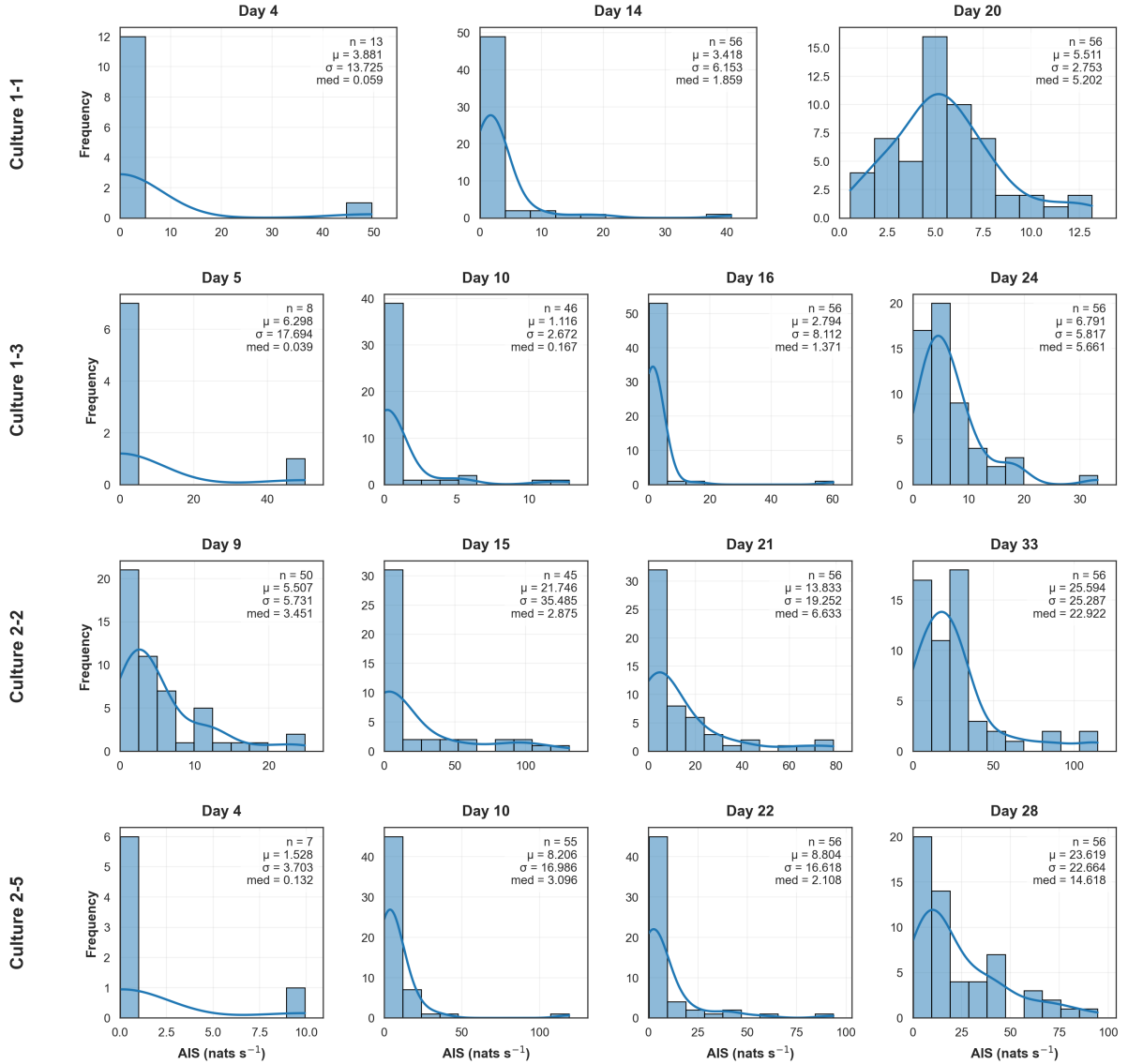


FIGURE 5.2: Developmental evolution of AIS histograms across cultures. Early stages show sparse, low-AIS distributions, while later stages broaden and shift right as more electrodes store memory (n indicates active electrodes).

5.3.3 Developmental Characteristics and Heterogeneity of AIS Spatial Distribution

From a spatial distribution perspective, we further analyzed the spatiotemporal evolution of AIS across electrodes through Figure 5.3. In this figure, each circle represents the location of an electrode, arranged according to the original experimental electrode layout to visually demonstrate the spatial positions of AIS within the network. Electrodes that did not reach statistical significance are marked in grey, while significant electrodes are represented by darker colors and larger circles indicating stronger AIS.

At the early stage (DIV 4–5), we observed a few electrodes showing extremely high AIS values, termed "pioneer electrodes," in Culture 1-1 and Culture 1-3. These pioneer electrodes did not exhibit any clear spatial pattern. As the network matured, although these early pioneer electrodes continued to show significant AIS, their AIS values did not remain consistently high, gradually settling at moderate intensity during the intermediate and late stages.

Overall, the spatial development pattern of AIS did not present a clear and uniform regularity; each culture exhibited distinct patterns of electrode activity. For example, Culture 2-2 displayed some electrodes with extremely high AIS values during later stages, whereas other cultures showed relatively uniform AIS spatial distributions. This highlights significant heterogeneity among different cultures even at identical developmental stages: some cultures exhibited locally concentrated AIS peaks, while others demonstrated a more uniformly distributed network memory structure.

5.3.4 AIS Early Lock-in Phenomenon

We further explored whether AIS exhibits a phenomenon similar to the "early lock-in" observed in TE, specifically whether early-stage AIS distributions remain highly correlated with subsequent developmental stages. To investigate this issue, we computed and plotted correlation matrices of AIS spatial distributions between different recording days within each culture (see Appendix A.2). Due to the presence of extremely high outliers in AIS distributions, we employed Spearman correlation coefficients to effectively reduce the impact of extreme values and more reliably reveal rank correlations of AIS distributions across different time points.

The results showed that culture 2-2 exhibited consistently significant high correlations across all pairs of recording days: for instance, the Spearman correlation between DIV 9–15 reached as high as 0.71, and even over the longer span between DIV 9–33, it maintained a correlation of 0.48 (Figure 5.4). This suggests that the spatial structure of AIS in this culture was determined very early and remained stable thereafter. In other cultures, the recording day pairs began earlier, and correlations at early pairs (such as DIV 4–10) did not show significant correlations yet. However, notably, correlations significantly increased during transitions from intermediate to late stages, peaking around 0.65, indicating these cultures' AIS spatial patterns also gradually locked in during the intermediate stage.

The overall correlation matrix plots for all cultures (see Figure 5.5) clearly show highly significant correlations primarily along the diagonal positions between adjacent recording days, with weaker or

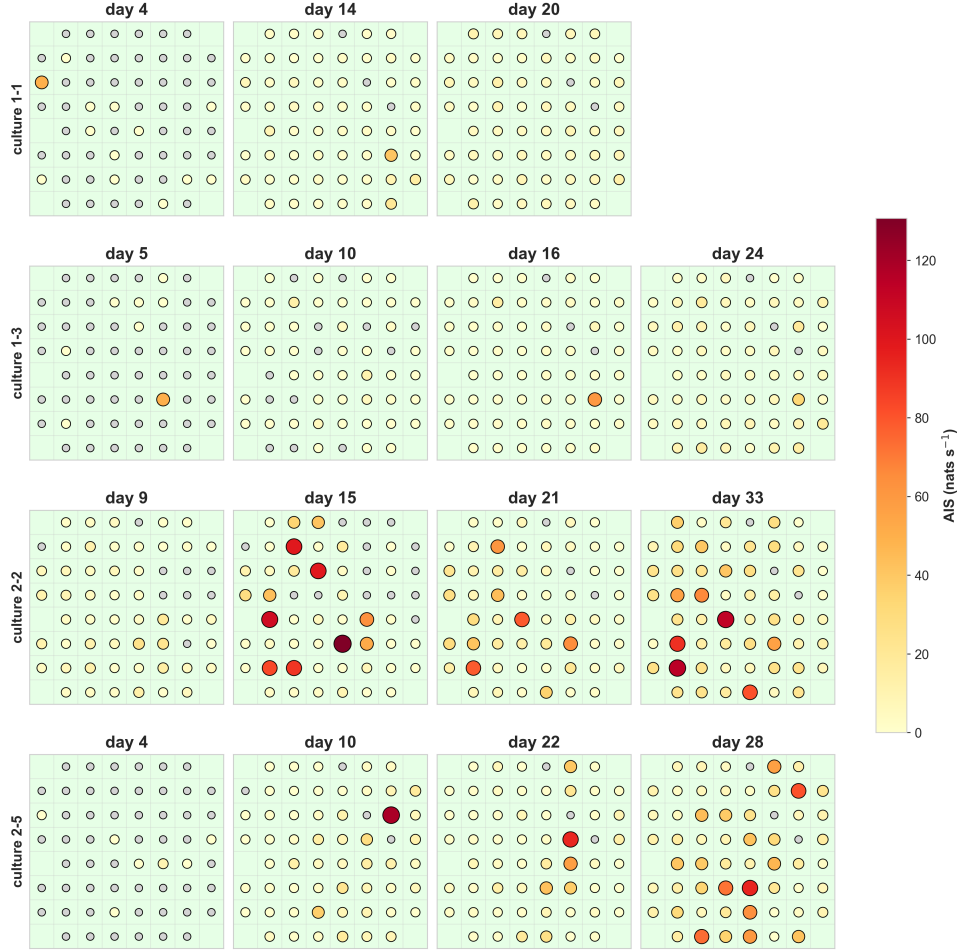


FIGURE 5.3: Spatial-temporal development of AIS across electrode arrays for all four cultures. Each panel shows the spatial distribution at different developmental stages. Gray circles indicate non-significant AIS values, while colored circles show significant ones. The progression demonstrates spatial expansion from sparse early activation to widespread mature network activation.

insignificant correlations only observed between the earliest DIV 4–10 recordings. This further supports the "early lock-in" phenomenon of AIS spatial patterns, gradually stabilizing from early to intermediate developmental stages and persisting into later stages.

5.4 Discussion and Conclusion

In this study, we discovered that critical structural features of information storage become "locked-in" at the early stages of neural development. As detailed in Section 5.3.4, significant correlations between recording days emerged during intermediate stage consistent with (Shorten et al., 2022). The heatmap of

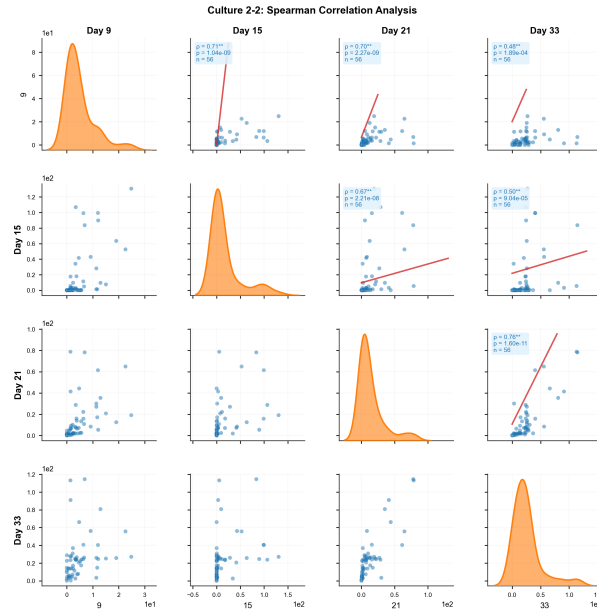


FIGURE 5.4: AIS spatial correlation matrix for Culture 2-2 across DIV 9, 15, 21, and 33. This culture demonstrates significant correlations between early developmental stages and persistent correlations across extended periods.

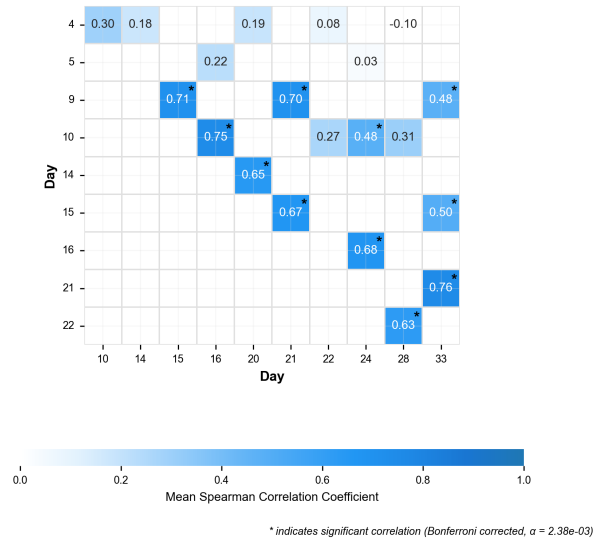


FIGURE 5.5: Temporal correlation structure of AIS patterns across developmental stages. Strong correlation blocks indicate memory pattern stabilization, while weak correlations reflect dynamic reorganization phases.

correlation matrices presented in Figure 5.5 reveals high correlation coefficients, further supporting the phenomenon of "early lock-in" and demonstrating sustained high correlation between early and later developmental patterns. This early lock-in could explain a fundamental developmental framework within

neural networks: once certain activity patterns or connection structures are established during early developmental or learning phases, they become "locked-in," making them resistant to change (Knudsen, 2004). This phenomenon profoundly influences the encoding and retrieval of memory, and reflects the self-organizing nature of neural networks.

Existing models indicate that, in *in vitro* cultured neural networks (such as MEAs), a small subset of early-maturing neurons, characterized by intrinsic bursting tendencies—often termed "leader" neurons—typically fire first and initiate synchronized network bursts (Picardo et al., 2011). Additionally, early-generated GABAergic leader neurons, born during the embryonic stage, have been observed to display early "leader" firing patterns and can drive entire network activity from single cells. Therefore, electrodes showing early high AIS signals in our study might correspond to these highly excitable pacemaker neurons. However, further experiments are required to validate whether neurons exhibiting the earliest firing indeed corresponds to those demonstrating high AIS.

During the intermediate stage of neural network development, the overall AIS levels increased more gradually, with some culture samples exhibiting no significant rise. Notably, other researchers analyzing the same publicly available dataset found that information transfer capability, measured using Transfer Entropy (TE), significantly lags behind AIS maturation (Shorten et al., 2022). Specifically, in the early stage (DIV 4–5), although the median AIS values were extremely low overall, individual electrodes demonstrated relatively high AIS, while TE values remained near zero. Entering the intermediate stage, AIS rapidly increased to approximately 3.08 nats s^{-1} , whereas TE remained below 0.06 bits s^{-1} . Only in the late stage (after DIV 14), when AIS increased more gradually, did TE significantly rise. This early maturation of AIS, followed by a slow grow period, provides a critical "time window" for subsequent TE development (Grubb and Burrone, 2010), further supporting the sequential maturation model: first intrinsic memory formation followed by network information transfer capability. These data strongly support a sequential maturation model—initial single-neuron memory followed by large-scale information transmission.

In summary, these findings establish AIS as a powerful and conserved metric for tracking neural network development. By bridging neurobiological principles with computational modeling, the insights derived from AIS analyses hold significant potential for informing therapeutic strategies and enhancing our understanding of neural network maturation.

Conclusions

This thesis introduced and validated a novel continuous-time estimator for Active Information Storage (AIS) in neural spike trains, with the goal of advancing our understanding of memory development in neural networks. By developing a framework that operates directly on spike timing data—thus avoiding arbitrary temporal binning—we preserved the fine temporal structure of neural activity. Applying this estimator to neural culture recordings, we discovered fundamental principles governing memory development in biological neural networks. Our findings reveal that memory capabilities increase markedly during development through distinct phases: from sparse, localized activation in "pioneer electrodes" to widespread, distributed temporal processing across the entire network. Critically, we identified an "early lock-in" phenomenon where memory patterns stabilize around DIV 14–16, and established a sequential maturation model where internal memory develops before information transfer capabilities. These results establish AIS as a valuable complement to existing information-theoretic tools, offering unique insights into the temporal dynamics of neural network maturation and the fundamental relationship between memory and communication in biological systems.

6.1 Summary of Contributions

The primary contributions of this thesis are threefold. First, I developed and implemented a robust computational framework for continuous-time AIS estimation tailored to neural spike train analysis, bridging the gap between theoretical concepts and empirical application. Second, I rigorously validated this estimator using synthetic datasets with known ground-truth properties, demonstrating its accuracy and reliability across a range of firing patterns. Third, I applied the validated framework to real neural culture data, uncovering new insights into memory development during neural network maturation.

The results are organized into two main areas: validation on synthetic data and analysis of neural culture development, each providing critical evidence for the estimator's performance and biological relevance.

6.1.1 Synthetic Data Validation

The continuous-time AIS estimator performed reliably across diverse synthetic point process models. For memoryless processes, it correctly converged to the theoretical AIS value of zero, confirming its ability to identify systems without memory. In refractory processes, the estimator closely matched theoretical predictions, with relative errors under 2%, demonstrating sensitivity to short-term temporal dependencies. For bursting processes, performance improved with higher firing rates, with relative errors decreasing from 13.8% to 3%, successfully capturing complex temporal patterns. The estimator also exhibited rapid convergence, requiring only about 200 spikes for stable estimates, and sampling variance decreased as data volume increased. Collectively, these results confirm the estimator's accuracy and efficiency, supporting its application to real neural data.

6.1.2 Neural Culture Network Development Analysis

Applying the estimator to four neural culture networks, we found a clear, robust increase in memory capacity (AIS) as networks matured. Median AIS values rose from near zero in early development (DIV 4–5) to 5–22 nats s^{-1} in mature stages (DIV 24–33), with statistical tests confirming this trend. Early development featured sparse memory, with only a few "pioneer electrodes" showing high AIS, while later stages saw widespread, network-wide memory emergence.

A key finding was the "early lock-in" of AIS pattern. Notably, it can be observed that AIS maturation consistently preceded the growth of information transfer (TE), suggesting the possibility a sequential model where internal memory develops before network communication.

Overall, these results highlight AIS as a sensitive biomarker for neural development, revealing that memory organization consolidates early and underpins later network communication.

6.2 Limitations

While this study offers important insights, several limitations should be noted to clarify the scope of our conclusions and guide future research.

Methodologically, we used only embedding vectors of length 1 ($\ell = 1$), which captures immediate inter-spike interval dependencies but may underestimate more complex, long-term memory dynamics in

mature neural networks. Additionally, our estimator employed fixed parameters (nearest-neighbor value $k = 4$ and Chebyshev distance metric) without systematic exploration of alternative settings or distance measures. Although these choices were validated on synthetic data, different parameter configurations or embedding lengths could be more suitable for specific neural activity patterns. Furthermore, our implementation did not incorporate exclusion windows around spike events, which could potentially introduce edge effects or artifacts in the entropy estimation, particularly for neurons with very short inter-spike intervals or during high-frequency bursting activity.

Experimentally, our analysis was limited by sparse temporal sampling (3-4 recording days per culture) and a small sample size (four rat cortical culture networks). This restricts our ability to capture rapid developmental transitions and may not reflect the full variability across preparations, species, brain regions, or developmental contexts. Our focus on rat cortical cultures further limits generalizability to other systems.

Finally, our implementation assumes stationarity within each recording session, which may not hold during periods of rapid developmental change. Short recording durations and the dynamic nature of developing neural networks present challenges for analyses based on stable statistical properties, underscoring the need for methodological advances that can accommodate non-stationary dynamics.

In summary, limitations in embedding strategy, parameter selection, sampling frequency, sample size, and assumptions of stationarity constrain the generalizability and precision of our findings. Addressing these issues in future work will enable a more comprehensive and robust understanding of neural memory development.

6.3 Future Work

Building on this research, future work can focus on methodological enhancements and experimental validation to deepen the understanding of neural memory development.

Implementing variable embedding lengths ($\ell = 2, 3, 5$) alongside adaptive time window techniques will allow the capture of longer-term memory patterns and effectively handle non-stationary dynamics during critical developmental transitions. The discovery of the sequential maturation model—where AIS develops before TE—provides a promising direction for investigating the fundamental relationship between internal memory and information transfer. Combined AIS-TE analyses could clarify the temporal

coupling between memory consolidation and communication emergence, uncover patterns of resource allocation during the critical DIV 14–16 transition period, explore potential trade-offs between internal processing and cross-regional communication, and validate the sequential emergence hypothesis across different neural systems. Our findings showing that AIS reaches substantial values ($1\text{--}3 \text{ nats s}^{-1}$) while TE remains near zero, followed by TE stabilization only after AIS plateau, suggest a fundamental organizational principle that warrants further investigation.

The identification of "pioneer electrodes" opens important avenues for targeted experimental validation. Future studies could employ cell-type-specific markers and optogenetic manipulations to test whether these high-AIS electrodes correspond to early-maturing pacemaker neurons that drive network bursts. Investigating the causal role of pioneer circuits in network-wide memory development through selective stimulation or silencing experiments could reveal fundamental mechanisms of neural network self-organization.

The "early lock-in" phenomenon discovered around DIV 14–16, where spatial correlations jump from below 0.3 to above 0.6, represents a critical developmental window that merits intensive study. High-frequency daily sampling during DIV 12–18 could precisely capture the dynamics of memory pattern stabilization and identify the molecular and synaptic mechanisms underlying this transition. Understanding what triggers and maintains this lock-in could have profound implications for critical period plasticity and therapeutic interventions.

Additionally, systematically evaluating multiple nearest-neighbor (k) values would optimize AIS estimation by capturing a broader range of data structures and temporal dynamics. Smaller k values, while sensitive to fine-scale temporal dynamics, are prone to noise and outliers. Conversely, larger k values reduce noise but might miss subtle temporal patterns (Singh and Póczos (2016); Fournier and Delatre (2016)). Exploring various k values thus promises enhanced robustness and generalizability of AIS analyses.

Extending this analytical framework to human neural organoids and in vivo recordings would enhance its broader applicability and translational potential. A recent study by (Kapucu et al., 2022) provided an open-access dataset of microelectrode array recordings, comparing the functional maturation of human pluripotent stem cell (hPSC)-derived neuronal networks with rat embryonic cortical cultures. They observed spontaneous firing, burst dynamics, and network synchronization across DIV 3–77 for human

networks and DIV 2–35 for rat networks, revealing species-specific maturation trajectories and electrophysiological signatures absent from rodent models (Kapucu et al., 2022). Future research could benefit from incorporating analogous hPSC-derived MEA datasets to achieve more physiologically relevant insights into memory development and information storage. This approach could uncover human-specific developmental dynamics, test whether the sequential maturation model and early lock-in phenomena are conserved across species, and significantly enhance the clinical relevance and applicability of continuous-time AIS analyses.

Together, these future research directions aim to optimize methodological approaches, expand biological insights, and ultimately advance our understanding of neural computation and developmental mechanisms, particularly the fundamental relationship between memory, communication, and network self-organization.

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Additional Results and Extended Analysis

This appendix presents additional experimental results, extended statistical analyses, and supplementary findings that support and complement the main results presented in the thesis. The appendix is organized into two main sections: (1) comprehensive statistical analyses including pairwise comparisons and descriptive statistics, and (2) detailed spatial correlation analyses showing temporal evolution of memory patterns across all neural cultures.

A.1 Comprehensive Statistical Analysis

This section provides detailed statistical analyses that extend beyond the main results presented in Chapter 5, including complete pairwise comparisons and descriptive statistics for all neural cultures.

A.1.1 Pairwise Wilcoxon Test Results

Table A.1 presents the complete statistical analysis of pairwise Wilcoxon tests comparing median AIS values across developmental stages for all four neural cultures. These results support the findings presented in Chapter 5 regarding the dramatic increases in memory utilization during neural development.

A.1.2 Descriptive Statistics and Variability Analysis

Table A.2 presents the complete summary of mean AIS values, standard deviations, and electrode counts for all neural cultures across developmental stages. These results complement the median-based analysis presented in the main text, showing similar developmental trends while highlighting the high variability in AIS values due to extreme outliers at individual electrodes.

TABLE A.1: Pairwise Wilcoxon tests for median AIS differences across developmental stages. All significant comparisons after Bonferroni correction ($\alpha = 0.00238$) are shown.

Culture	Early DIV	Late DIV	Median Early	Median Late	p-value
1-1	4	14	0.00	1.86	1.11×10^{-9}
1-1	4	20	0.00	5.20	2.59×10^{-10}
1-1	14	20	1.86	5.20	9.15×10^{-8}
1-3	5	10	0.00	0.11	6.97×10^{-9}
1-3	5	16	0.00	1.37	1.36×10^{-9}
1-3	5	24	0.00	5.66	3.19×10^{-10}
1-3	10	16	0.11	1.37	6.97×10^{-8}
1-3	10	24	0.11	5.66	1.29×10^{-10}
1-3	16	24	1.37	5.66	1.16×10^{-9}
2-2	9	21	3.06	6.63	8.76×10^{-6}
2-2	9	33	3.06	22.92	3.55×10^{-10}
2-2	15	33	1.14	22.92	2.13×10^{-7}
2-2	21	33	6.63	22.92	1.03×10^{-8}
2-5	4	10	0.00	3.08	7.55×10^{-11}
2-5	4	22	0.00	2.11	7.55×10^{-11}
2-5	4	28	0.00	14.62	7.55×10^{-11}
2-5	10	28	3.08	14.62	2.13×10^{-7}
2-5	22	28	2.11	14.62	2.03×10^{-9}

TABLE A.2: Mean AIS values, standard deviations, and electrode counts across developmental stages for all neural cultures. High standard deviations reflect the presence of extreme outlier electrodes with exceptionally high AIS values.

Culture	DIV	Mean AIS (nats s ⁻¹)	Median AIS (nats s ⁻¹)	Std Dev	Electrode Count
1-1	4	0.900	0.001	6.621	56
1-1	14	3.418	1.859	6.153	56
1-1	20	5.511	5.202	2.753	56
1-3	5	0.899	0.001	6.693	56
1-3	10	0.917	0.114	2.455	56
1-3	16	2.794	1.371	8.112	56
1-3	24	6.791	5.661	5.817	56
2-2	9	4.917	3.056	5.676	56
2-2	15	17.476	1.138	32.913	56
2-2	21	13.833	6.633	19.252	56
2-2	33	25.594	22.922	25.287	56
2-5	4	0.190	-0.000	1.325	56
2-5	10	8.060	3.078	16.867	56
2-5	22	8.804	2.108	16.618	56
2-5	28	23.619	14.618	22.664	56

The mean values generally follow the same developmental trends as the median values, with consistent increases from early to late developmental stages. However, the large standard deviations (often exceeding the mean values themselves) highlight the presence of extreme outlier electrodes with exceptionally high AIS values. This high variability justifies the use of median-based analysis in the main text, as medians provide more robust estimates of central tendency in the presence of such outliers.

A.2 Spatial Correlation Analysis

This section presents detailed correlation matrices of AIS spatial distributions across different recording days for each of the four neural cultures analyzed in Chapter 5. These matrices illustrate the temporal evolution and stability of memory organization patterns throughout neural development.

Figure A.1 shows the correlation matrix for Culture 1-1 across three developmental stages (DIV 4, 14, and 20).

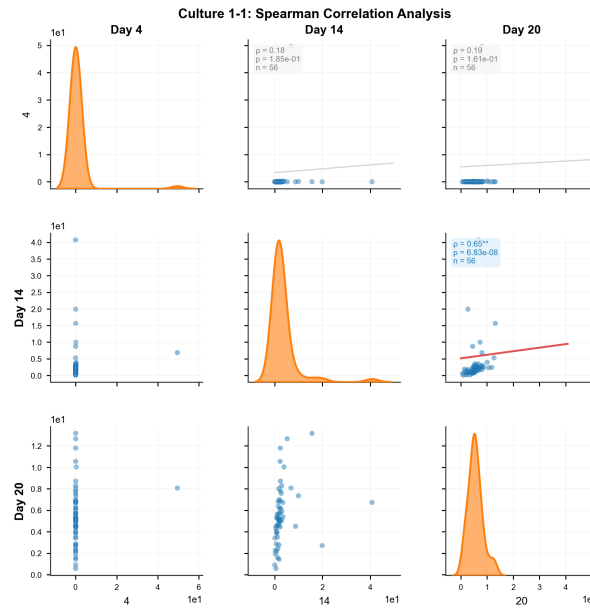


FIGURE A.1: AIS spatial correlation matrix for Culture 1-1 across DIV 4, 14, and 20.

Figure A.2 presents the correlation matrix for Culture 1-3 across four developmental stages (DIV 5, 10, 16, and 24).

Figure A.3 presents the correlation matrix for Culture 2-5 across four developmental stages (DIV 4, 10, 22, and 28).

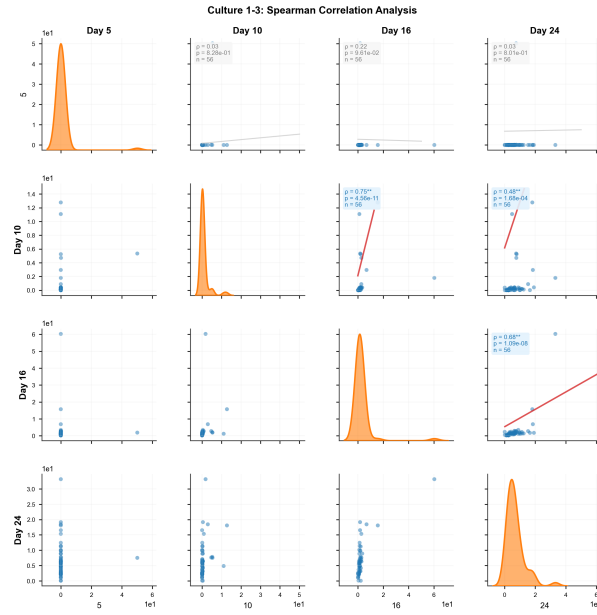


FIGURE A.2: AIS spatial correlation matrix for Culture 1-3 across DIV 5, 10, 16, and 24.

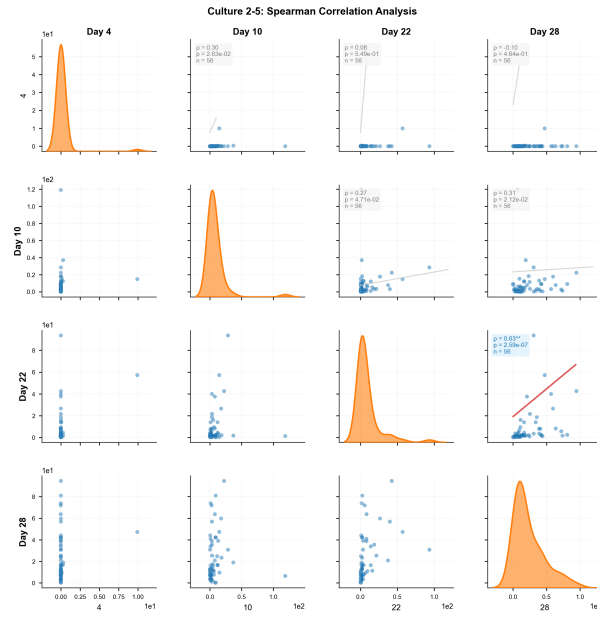


FIGURE A.3: AIS spatial correlation matrix for Culture 2-5 across DIV 4, 10, 22, and 28.